Neural Computation -- 2002

137 article(s), all issues

1. Self-organization in the basal ganglia with modulation of reinforcement signals

NakaharaHiroyuki, AmariShun-ichi, HikosakaOkihide

(no DOI)

Self-organization is one of fundamental brain computations for forming efficient representations of information. Experimental support for this idea has been largely limited to the developmental and...

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2. Bayesian model assessment and comparison using cross-validation predictive densities

VehtariAki, LampinenJouko

https://doi.org/10.5555/641610.641617

In this work, we discuss practical methods for the assessment, comparison, and selection of complex hierarchical Bayesian models. A natural way to assess the goodness of the model is to estimate it...

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3. (untitled)

(authors unknown)

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14. (untitled)

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No abstract available.

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15. Mechanisms Shaping Fast Excitatory Postsynaptic Currents in the Central Nervous System

M.I. Glavinovi■

https://doi.org/10.1162/089976602753284437 | pp. 1-19

How different factors contribute to determine the time course of the basic element of fast glutamate-mediated excitatory postsynaptic currents (mEPSCs) in the central nervous system has been a focus of interest of neurobiologists for some years. In spite of intensive investigations, these mechanisms are not well understood. In this review, basic hypotheses are summarized, and a new hypothesis is proposed, which holds that desensitization of AMPA receptors plays a major role in shaping the time course of fast mEPSCs. According to the new hypothesis, desensitization shortens the time course of mEPSCs largely by reducing the buffering of glutamate molecules by AMPA receptors. The hypothesis accounts for numerous findings on fast mEPSCs and is expected to be equally fruitful as a framework for further experimental and theoretical investigations.

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16. Adjusting the Outputs of a Classifier to New aPrioriProbabilities: A Simple Procedure

Marco Saerens, Patrice Latinne, Christine Decaestecker

https://doi.org/10.1162/089976602753284446 | pp. 21-41

It sometimes happens (for instance in case control studies) that a classifier is trained on a data set that does not reflect the true a priori probabilities of the target classes on real-world data. This may have a negative effect on the classification accuracy obtained on the real-world data set, especially when the classifier's decisions are based on the a posteriori probabilities of class membership. Indeed, in this case, the trained classifier provides estimates of the a posteriori probabilities that are not valid for this real-world data set (they rely on the a priori probabilities of the training set). Applying the classifier as is (without correcting its outputs with respect to these new conditions) on this new data set may thus be suboptimal. In this note, we present a simple iterative procedure for adjusting the outputs of the trained classifier with respect to these new a priori probabilities without having to refit the model, even when these probabilities are not known in advance. As a by-product, estimates of the new a priori probabilities are also obtained. This iterative

algorithm is a straightforward instance of the expectation-maximization (EM) algorithm and is shown to maximize the likelihood of the new data. Thereafter, we discuss a statistical test that can be applied to decide if the a priori class probabilities have changed from the training set to the real-world data. The procedure is illustrated on different classification problems involving a multilayer neural network, and comparisons with a standard procedure for a priori probability estimation are provided. Our original method, based on the EM algorithm, is shown to be superior to the standard one for a priori probability estimation. Experimental results also indicate that the classifier with adjusted outputs always performs better than the original one in terms of classification accuracy, when the a priori probability conditions differ from the training set to the real-world data. The gain in classification accuracy can be significant.

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17. Unitary Events in Multiple Single-Neuron Spiking Activity: I. Detection and Significance

Sonja Grün, Markus Diesmann, Ad Aertsen

https://doi.org/10.1162/089976602753284455 | pp. 43-80

It has been proposed that cortical neurons organize dynamically into functional groups (cell assemblies) by the temporal structure of their joint spiking activity. Here, we describe a novel method to detect conspicuous patterns of coincident joint spike activity among simultaneously recorded single neurons. The statistical significance of these unitary events of coincident joint spike activity is evaluated by the joint-surprise. The method is tested and calibrated on the basis of simulated, stationary spike trains of independently firing neurons, into which coincident joint spike events were inserted under controlled conditions. The sensitivity and specificity of the method are investigated for their dependence on physiological parameters (firing rate, coincidence precision, coincidence pattern complexity) and temporal resolution of the analysis. In the companion article in this issue, we describe an extension of the method, designed to deal with nonstationary firing rates.

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18. Unitary Events in Multiple Single-Neuron Spiking Activity: II. Nonstationary Data

Sonja Grün, Markus Diesmann, Ad Aertsen

https://doi.org/10.1162/089976602753284464 | pp. 81-119

In order to detect members of a functional group (cell assembly) in simultaneously recorded neuronal spiking activity, we adopted the widely used operational definition that membership in a common assembly is expressed in near-simultaneous spike activity. Unitary event analysis, a statistical method to detect the significant occurrence of coincident spiking activity in stationary data, was recently developed (see the companion article in this issue). The technique for the detection of unitary events is based on the assumption that the underlying processes are stationary in time. This requirement, however, is usually not fulfilled in neuronal data. Here we describe a method that properly normalizes for changes of rate: the unitary events by moving window analysis (UEMWA). Analysis for unitary events is performed separately in overlapping time segments by sliding a window of constant width along the data. In each window, stationarity is assumed. Performance and sensitivity are demonstrated by use of simulated spike trains of independently firing neurons, into which coincident events are inserted. If cortical neurons organize dynamically into functional groups, the occurrence of near-simultaneous spike activity should be time

varying and related to behavior and stimuli. UEMWA also accounts for these potentially interesting nonstationarities and allows locating them in time. The potential of the new method is illustrated by results from multiple single-unit recordings from frontal and motor cortical areas in awake, behaving monkey.

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19. Statistical Significance of Coincident Spikes: Count-Based Versus Rate-Based Statistics

Robert Gütig, Ad Aertsen, Stefan Rotter

https://doi.org/10.1162/089976602753284473 | pp. 121-153

Inspired by different conceptualizations of temporal neural coding schemes, there has been recent interest in the search for signs of precisely synchronized neural activity in the cortex. One method developed for this task is unitary-event analysis. This method tests multiple single neuron recordings for short epochs with significantly more coincident spikes than expected from independent neurons. We reformulated the statistical test underlying this method using a coincidence count distribution based on empirical spike counts rather than on estimated spike probabilities. In the case of two neurons, the requirement of stationary firing rates, originally imposed on both neurons, can be relaxed; only the rate of one neuron needs to be stationary, while the other may follow an arbitrary time course. By analytical calculations of the test power curves of the original and the revised method, we demonstrate that the test power can be increased by a factor of two or more in physiologically realistic regimes. In addition, we analyze the effective significance levels of both methods for neural firing rates ranging between 0.2 Hz and 30 Hz.

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20. Representational Accuracy of Stochastic Neural Populations

Stefan Wilke, Christian W. Eurich

https://doi.org/10.1162/089976602753284482 | pp. 155-189

Fisher information is used to analyze the accuracy with which a neural population encodes D stimulus features. It turns out that the form of response variability has a major impact on the encoding capacity and therefore plays an important role in the selection of an appropriate neural model. In particular, in the presence of baseline firing, the reconstruction error rapidly increases with D in the case of Poissonian noise but not for additive noise. The existence of limited-range correlations of the type found in cortical tissue yields a saturation of the Fisher information content as a function of the population size only for an additive noise model. We also show that random variability in the correlation coefficient within a neural population, as found empirically, considerably improves the average encoding quality. Finally, the representational accuracy of populations with inhomogeneous tuning properties, either with variability in the tuning widths or fragmented into specialized subpopulations, is superior to the case of identical and radially symmetric tuning curves usually considered in the literature.

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21. Supervised Dimension Reduction of Intrinsically Low-Dimensional Data

Nikos Vlassis, Yoichi Motomura, Ben Kröse

https://doi.org/10.1162/089976602753284491 | pp. 191-215

High-dimensional data generated by a system with limited degrees of freedom are often constrained in low-dimensional manifolds in the original space. In this article, we investigate dimension-reduction methods for such intrinsically low-dimensional data through linear projections that preserve the manifold structure of the data. For intrinsically one dimensional data, this implies projecting to a curve on the plane with as few intersections as possible. We are proposing a supervised projection pursuit method that can be regarded as an extension of the single-index model for nonparametric regression. We show results from a toy and two robotic applications.

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22. Clustering Based on Conditional Distributions in an Auxiliary Space

Janne Sinkkonen, Samuel Kaski

https://doi.org/10.1162/089976602753284509 | pp. 217-239

We study the problem of learning groups or categories that are local in the continuous primary space but homogeneous by the distributions of an associated auxiliary random variable over a discrete auxiliary space. Assuming that variation in the auxiliary space is meaningful, categories will emphasize similarly meaningful aspects of the primary space. From a data set consisting of pairs of primary and auxiliary items, the categories are learned by minimizing a Kullback-Leibler divergence-based distortion between (implicitly estimated) distributions of the auxiliary data, conditioned on the primary data. Still, the categories are defined in terms of the primary space. An online algorithm resembling the traditional Hebb-type competitive learning is introduced for learning the categories. Minimizing the distortion criterion turns out to be equivalent to maximizing the mutual information between the categories and the auxiliary data. In addition, connections to density estimation and to the distributional clustering paradigm are outlined. The method is demonstrated by clustering yeast gene expression data from DNA chips, with biological knowledge about the functional classes of the genes as the auxiliary data.

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23. On the Complexity of Computing and Learning with Multiplicative Neural Networks

Michael Schmitt

https://doi.org/10.1162/08997660252741121 | pp. 241-301

In a great variety of neuron models, neural inputs are combined using the summing operation. We introduce the concept of multiplicative neural networks that contain units that multiply their inputs instead of summing them and thus allow inputs to interact nonlinearly. The class of multiplicative neural networks comprises such widely known and well-studied network types as higher-order networks and product unit networks. We investigate the complexity of computing and learning for multiplicative neural networks. In particular, we derive upper and lower bounds on the Vapnik-Chervonenkis (VC) dimension and the pseudo-dimension for various types of networks with multiplicative units. As the most general case, we consider feedforward networks consisting of product and sigmoidal units, showing that their pseudo-dimension is bounded from above by a polynomial with the same order of magnitude as the currently best-known bound for purely sigmoidal networks. Moreover, we show that this bound holds even when the unit type, product or sigmoidal, may be learned. Crucial for these results are calculations of solution set components bounds for new network classes. As to lower bounds, we construct product unit networks of fixed depth with super-linear VC dimension. For sigmoidal networks of higher order, we establish polynomial bounds that, in contrast to

previous results, do not involve any restriction of the network order. We further consider various classes of higher-order units, also known as sigma-pi units, that are characterized by connectivity constraints. In terms of these, we derive some asymptotically tight bounds. Multiplication plays an important role in both neural modeling of biological behavior and computing and learning with artificial neural networks. We briefly survey research in biology and in applications where multiplication is considered an essential computational element. The results we present here provide new tools for assessing the impact of multiplication on the computational power and the learning capabilities of neural networks.

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24. A Lagrange Multiplier and Hopfield-Type Barrier Function Method for the Traveling Salesman Problem

Chuangyin Dang, Lei Xu

https://doi.org/10.1162/08997660252741130 | pp. 303-324

A Lagrange multiplier and Hopfield-type barrier function method is proposed for approximating a solution of the traveling salesman problem. The method is derived from applications of Lagrange multipliers and a Hopfield-type barrier function and attempts to produce a solution of high quality by generating a minimum point of a barrier problem for a sequence of descending values of the barrier parameter. For any given value of the barrier parameter, the method searches for a minimum point of the barrier problem in a feasible descent direction, which has a desired property that lower and upper bounds on variables are always satisfied automatically if the step length is a number between zero and one. At each iteration, the feasible descent direction is found by updating Lagrange multipliers with a globally convergent iterative procedure. For any given value of the barrier parameter, the method converges to a stationary point of the barrier problem without any condition on the objective function. Theoretical and numerical results show that the method seems more effective and efficient than the softassign algorithm.

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25. The Time-Rescaling Theorem and Its Application to Neural Spike Train Data Analysis

Emery N. Brown, Riccardo Barbieri, Valérie Ventura, Robert E. Kass, Loren M. Frank

https://doi.org/10.1162/08997660252741149 | pp. 325-346

Measuring agreement between a statistical model and a spike train data series, that is, evaluating goodness of fit, is crucial for establishing the model's validity prior to using it to make inferences about a particular neural system. Assessing goodness-of-fit is a challenging problem for point process neural spike train models, especially for histogram-based models such as perstimulus time histograms (PSTH) and rate functions estimated by spike train smoothing. The time-rescaling theorem is a well-known result in probability theory, which states that any point process with an integrable conditional intensity function may be transformed into a Poisson process with unit rate. We describe how the theorem may be used to develop goodness-of-fit tests for both parametric and histogram-based point process models of neural spike trains. We apply these tests in two examples: a comparison of PSTH, inhomogeneous Poisson, and inhomogeneous Markov interval models of neural spike trains from the supplementary eye field of a macque monkey and a comparison of temporal and spatial smoothers, inhomogeneous Poisson, inhomogeneous gamma, and inhomogeneous inverse gaussian models of rat hippocampal place cell spiking activity. To help make the logic behind the time-rescaling theorem more accessible to researchers in neuroscience, we

present a proof using only elementary probability theory arguments. We also show how the theorem may be used to simulate a general point process model of a spike train. Our paradigm makes it possible to compare parametric and histogram-based neural spike train models directly. These results suggest that the time-rescaling theorem can be a valuable tool for neural spike train data analysis.

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26. The Impact of Spike Timing Variability on the Signal-Encoding Performance of Neural Spiking Models

Amit Manwani, Peter N. Steinmetz, Christof Koch

https://doi.org/10.1162/08997660252741158 | pp. 347-367

It remains unclear whether the variability of neuronal spike trains in vivo arises due to biological noise sources or represents highly precise encoding of temporally varying synaptic input signals. Determining the variability of spike timing can provide fundamental insights into the nature of strategies used in the brain to represent and transmit information in the form of discrete spike trains. In this study, we employ a signal estimation paradigm to determine how variability in spike timing affects encoding of random time-varying signals. We assess this for two types of spiking models: an integrate-and-fire model with random threshold and a more biophysically realistic stochastic ion channel model. Using the coding fraction and mutual information as information-theoretic measures, we quantify the efficacy of optimal linear decoding of random inputs from the model outputs and study the relationship between efficacy and variability in the output spike train. Our findings suggest that variability does not necessarily hinder signal decoding for the biophysically plausible encoders examined and that the functional role of spiking variability depends intimately on the nature of the encoder and the signal processing task; variability can either enhance or impede decoding performance.

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27. Temporal Correlations in Stochastic Networks of Spiking Neurons

Carsten Meyer, Carl van Vreeswijk

https://doi.org/10.1162/08997660252741167 | pp. 369-404

The determination of temporal and spatial correlations in neuronal activity is one of the most important neurophysiological tools to gain insight into the mechanisms of information processing in the brain. Its interpretation is complicated by the difficulty of disambiguating the effects of architecture, single-neuron properties, and network dynamics. We present a theory that describes the contribution of the network dynamics in a network of "spiking" neurons. For a simple neuron model including refractory properties, we calculate the temporal cross-correlations in a completely homogeneous, excitatory, fully connected network in a stable, stationary state, for stochastic dynamics in both discrete and continuous time. We show that even for this simple network architecture, the cross-correlations exhibit a large variety of qualitatively different properties, strongly dependent on the level of noise, the decay constant of the refractory function, and the network activity. At the critical point, the cross-correlations oscillate with a frequency that depends on the refractory properties or decay exponentially with a diverging damping constant (for "weak" refractory properties). We also investigate the effect of the synaptic time constants. It is shown that these time constants may, apart from their influence on the asymmetric peak arising from the direct synaptic connection, also affect the long-term properties of the cross-correlations.

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28. Measuring Information Spatial Densities

Michele Bezzi, Inés Samengo, Stefan Leutgeb, Sheri J. Y. Mizumori

https://doi.org/10.1162/08997660252741176 | pp. 405-420

A novel definition of the stimulus-specific information is presented, which is particularly useful when the stimuli constitute a continuous and metric set, as, for example, position in space. The approach allows one to build the spatial information distribution of a given neural response. The method is applied to the investigation of putative differences in the coding of position in hippocampus and lateral septum.

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29. Stochastic Trapping in a Solvable Model of On-Line Independent Component Analysis

Magnus Rattray

https://doi.org/10.1162/08997660252741185 | pp. 421-435

Previous analytical studies of on-line independent component analysis (ICA) learning rules have focused on asymptotic stability and efficiency. In practice, the transient stages of learning are often more significant in determining the success of an algorithm. This is demonstrated here with an analysis of a Hebbian ICA algorithm, which can find a small number of nongaussian components given data composed of a linear mixture of independent source signals. An idealized data model is considered in which the sources comprise a number of nongaussian and gaussian sources, and a solution to the dynamics is obtained in the limit where the number of gaussian sources is infinite. Previous stability results are confirmed by expanding around optimal fixed points, where a closed-form solution to the learning dynamics is obtained. However, stochastic effects are shown to stabilize otherwise unstable suboptimal fixed points. Conditions required to destabilize one such fixed point are obtained for the case of a single nongaussian component, indicating that the initial learning rate eta required to escape successfully is very low (eta = O(N(-2)) where N is the data dimension), resulting in very slow learning typically requiring O(N(3)) iterations. Simulations confirm that this picture holds for a finite system.

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30. A Neural-Network-Based Approach to the Double Traveling Salesman Problem

Alessio Plebe, A. M. Anile

https://doi.org/10.1162/08997660252741194 | pp. 437-471

The double traveling salesman problem is a variation of the basic traveling salesman problem where targets can be reached by two salespersons operating in parallel. The real problem addressed by this work concerns the optimization of the harvest sequence for the two independent arms of a fruit-harvesting robot. This application poses further constraints, like a collision-avoidance function. The proposed solution is based on a self-organizing map structure, initialized with as many artificial neurons as the number of targets to be reached. One of the key components of the process is the combination of competitive relaxation with a mechanism for deleting and creating artificial neurons. Moreover, in the competitive relaxation process, information about the trajectory connecting the neurons is combined with the distance of neurons from the target. This strategy prevents tangles in the trajectory and collisions between the two tours. Results of tests indicate that the proposed approach is efficient and reliable for harvest sequence planning. Moreover, the

enhancements added to the pure self-organizing map concept are of wider importance, as proved by a traveling salesman problem version of the program, simplified from the double version for comparison.

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31. What Geometric Visual Hallucinations Tell Us about the Visual Cortex

Paul C. Bressloff, Jack D. Cowan, Martin Golubitsky, Peter J. Thomas, Matthew C. Wiener

https://doi.org/10.1162/089976602317250861 | pp. 473-491

Many observers see geometric visual hallucinations after taking hallucinogens such as LSD, cannabis, mescaline or psilocybin; on viewing bright flickering lights; on waking up or falling asleep; in "near-death" experiences; and in many other syndromes. Klüver organized the images into four groups called form constants: (I) tunnels and funnels, (II) spirals, (III) lattices, including honeycombs and triangles, and (IV) cobwebs. In most cases, the images are seen in both eyes and move with them. We interpret this to mean that they are generated in the brain. Here, we summarize a theory of their origin in visual cortex (area V1), based on the assumption that the form of the retino-cortical map and the architecture of V1 determine their geometry. (A much longer and more detailed mathematical version has been published in Philosophical Transactions of the Royal Society B, 356 [2001].) We model V1 as the continuum limit of a lattice of interconnected hypercolumns, each comprising a number of interconnected iso-orientation columns. Based on anatomical evidence, we assume that the lateral connectivity between hypercolumns exhibits symmetries, rendering it invariant under the action of the Euclidean group E(2), composed of reflections and translations in the plane, and a (novel) shift-twist action. Using this symmetry, we show that the various patterns of activity that spontaneously emerge when V1's spatially uniform resting state becomes unstable correspond to the form constants when transformed to the visual field using the retino-cortical map. The results are sensitive to the detailed specification of the lateral connectivity and suggest that the cortical mechanisms that generate geometric visual hallucinations are closely related to those used to process edges, contours, surfaces, and textures.

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32. An Amplitude Equation Approach to Contextual Effects in Visual Cortex

Paul C. Bressloff, Jack D. Cowan

https://doi.org/10.1162/089976602317250870 | pp. 493-525

A mathematical theory of interacting hypercolumns in primary visual cortex (V1) is presented that incorporates details concerning the anisotropic nature of long-range lateral connections. Each hypercolumn is modeled as a ring of interacting excitatory and inhibitory neural populations with orientation preferences over the range 0 to 180 degrees. Analytical methods from bifurcation theory are used to derive nonlinear equations for the amplitude and phase of the population tuning curves in which the effective lateral interactions are linear in the amplitudes. These amplitude equations describe how mutual interactions between hypercolumns via lateral connections modify the response of each hypercolumn to modulated inputs from the lateral geniculate nucleus; such interactions form the basis of contextual effects. The coupled ring model is shown to reproduce a number of orientation-dependent and contrast-dependent features observed in center-surround experiments. A major prediction of the model is that the anisotropy in lateral connections results in a nonuniform modulatory effect of the surround that is correlated with the orientation of the center.

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33. Derivation of the Visual Contrast Response Function by Maximizing Information Rate

Allan Gottschalk

https://doi.org/10.1162/089976602317250889 | pp. 527-542

A graph of neural output as a function of the logarithm of stimulus intensity often produces an S-shaped function, which is frequently modeled by the hyperbolic ratio equation. The response of neurons in early vision to stimuli of varying contrast is an important example of this. Here, the hyperbolic ratio equation with a response exponent of two is derived exactly by considering the balance between information rate and the neural costs of making that information available, where neural costs are a function of synaptic strength and spike rate. The maximal response and semisaturation constant of the neuron can be related to the stimulus ensemble and therefore shift accordingly to exhibit contrast gain control and normalization.

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34. A Bayesian Framework for Sensory Adaptation

Norberto M. Grzywacz, Rosario M. Balboa

https://doi.org/10.1162/089976602317250898 | pp. 543-559

Adaptation allows biological sensory systems to adjust to variations in the environment and thus to deal better with them. In this article, we propose a general framework of sensory adaptation. The underlying principle of this framework is the setting of internal parameters of the system such that certain prespecified tasks can be performed optimally. Because sensorial inputs vary probabilistically with time and biological mechanisms have noise, the tasks could be performed incorrectly. We postulate that the goal of adaptation is to minimize the number of task errors. This minimization requires prior knowledge of the environment and of the limitations of the mechanisms processing the information. Because these processes are probabilistic, we formulate the minimization with a Bayesian approach. Application of this Bayesian framework to the retina is successful in accounting for a host of experimental findings.

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35. Analysis of Oscillations in a Reciprocally Inhibitory Network with Synaptic Depression

Adam L. Taylor, Garrison W. Cottrell, William B. Kristan

https://doi.org/10.1162/089976602317250906 | pp. 561-581

We present and analyze a model of a two-cell reciprocally inhibitory network that oscillates. The principal mechanism of oscillation is short-term synaptic depression. Using a simple model of depression and analyzing the system in certain limits, we can derive analytical expressions for various features of the oscillation, including the parameter regime in which stable oscillations occur, as well as the period and amplitude of these oscillations. These expressions are functions of three parameters: the time constant of depression, the synaptic strengths, and the amount of tonic excitation the cells receive. We compare our analytical results with the output of numerical simulations and obtain good agreement between the two. Based on our analysis, we conclude that the oscillations in our network are qualitatively different from those in networks that oscillate due to postinhibitory rebound, spike-frequency adaptation, or other intrinsic (rather than synaptic) adaptational mechanisms. In particular, our network can oscillate only via the synaptic escape mode of Skinner, Kopell, and Marder (1994).

36. Activity-Dependent Development of Axonal and Dendritic Delays, or, Why Synaptic Transmission Should Be Unreliable

Walter Senn, Martin Schneider, Berthold Ruf

https://doi.org/10.1162/089976602317250915 | pp. 583-619

Systematic temporal relations between single neuronal activities or population activities are ubiquitous in the brain. No experimental evidence, however, exists for a direct modification of neuronal delays during Hebbian-type stimulation protocols. We show that in fact an explicit delay adaptation is not needed if one assumes that the synaptic strengths are modified according to the recently observed temporally asymmetric learning rule with the downregulating branch dominating the upregulating branch. During development, slow, unbiased fluctuations in the transmission time, together with temporally correlated network activity, may control neural growth and implicitly induce drifts in the axonal delays and dendritic latencies. These delays and latencies become optimally tuned in the sense that the synaptic response tends to peak in the soma of the postsynaptic cell if this is most likely to fire. The nature of the selection process requires unreliable synapses in order to give successful synapses an evolutionary advantage over the others. The width of the learning function also determines the preferred dendritic delay and the preferred width of the postsynaptic response. Hence, it may implicitly determine whether a synaptic connection provides a precisely timed or a broadly tuned "contextual" signal.

37. Impact of Geometrical Structures on the Output of Neuronal Models: A Theoretical and Numerical Analysis

Jianfeng Feng, Guibin Li

https://doi.org/10.1162/089976602317250924 | pp. 621-640

What is the difference between the efferent spike train of a neuron with a large soma versus that of a neuron with a small soma? We propose an analytical method called the decoupling approach to tackle the problem. Two limiting cases-the soma is much smaller than the dendrite or vica versa-are theoretically investigated. For both the two-compartment integrate-and-fire model and Pinsky-Rinzel model, we show, both theoretically and numerically, that the smaller the soma is, the faster and the more irregularly the neuron fires. We further conclude, in terms of numerical simulations, that cells falling in between the two limiting cases form a continuum with respect to their firing properties (mean firing time and coefficient of variation of inter-spike intervals).

38. Sparse On-Line Gaussian Processes

Lehel Csató, Manfred Opper

https://doi.org/10.1162/089976602317250933 | pp. 641-668

We develop an approach for sparse representations of gaussian process (GP) models (which are Bayesian types of kernel machines) in order to overcome their limitations for large data sets. The method is based on a combination of a Bayesian on-line algorithm, together with a sequential construction of a relevant subsample of the data that fully specifies the prediction of the GP model. By using an appealing parameterization and projection techniques in a reproducing kernel Hilbert space, recursions for the

effective parameters and a sparse gaussian approximation of the posterior process are obtained. This allows for both a propagation of predictions and Bayesian error measures. The significance and robustness of our approach are demonstrated on a variety of experiments.

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39. Orthogonal Series Density Estimation and the Kernel Eigenvalue Problem

Mark Girolami

https://doi.org/10.1162/089976602317250942 | pp. 669-688

Kernel principal component analysis has been introduced as a method of extracting a set of orthonormal nonlinear features from multivariate data, and many impressive applications are being reported within the literature. This article presents the view that the eigenvalue decomposition of a kernel matrix can also provide the discrete expansion coefficients required for a nonparametric orthogonal series density estimator. In addition to providing novel insights into nonparametric density estimation, this article provides an intuitively appealing interpretation for the nonlinear features extracted from data using kernel principal component analysis.

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40. Natural Discriminant Analysis Using Interactive Potts Models

Jiann-Ming Wu

https://doi.org/10.1162/089976602317250951 | pp. 689-713

Natural discriminant analysis based on interactive Potts models is developed in this work. A generative model composed of piece-wise multivariate gaussian distributions is used to characterize the input space, exploring the embedded clustering and mixing structures and developing proper internal representations of input parameters. The maximization of a log-likelihood function measuring the fitness of all input parameters to the generative model, and the minimization of a design cost summing up square errors between posterior outputs and desired outputs constitutes a mathematical framework for discriminant analysis. We apply a hybrid of the mean-field annealing and the gradient-descent methods to the optimization of this framework and obtain multiple sets of interactive dynamics, which realize coupled Potts models for discriminant analysis. The new learning process is a whole process of component analysis, clustering analysis, and labeling analysis. Its major improvement compared to the radial basis function and the support vector machine is described by using some artificial examples and a real-world application to breast cancer diagnosis.

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41. Slow Feature Analysis: Unsupervised Learning of Invariances

Laurenz Wiskott, Terrence J. Sejnowski

https://doi.org/10.1162/089976602317318938 | pp. 715-770

Invariant features of temporally varying signals are useful for analysis and classification. Slow feature analysis (SFA) is a new method for learning invariant or slowly varying features from a vectorial input signal. It is based on a nonlinear expansion of the input signal and application of principal component analysis to this expanded signal and its time derivative. It is guaranteed to find the optimal solution within a family of functions directly and can learn to extract a large number of decorrelated features, which are ordered by

their degree of invariance. SFA can be applied hierarchically to process high-dimensional input signals and extract complex features. SFA is applied first to complex cell tuning properties based on simple cell output, including disparity and motion. Then more complicated input-output functions are learned by repeated application of SFA. Finally, a hierarchical network of SFA modules is presented as a simple model of the visual system. The same unstructured network can learn translation, size, rotation, contrast, or, to a lesser degree, illumination invariance for one-dimensional objects, depending on only the training stimulus. Surprisingly, only a few training objects suffice to achieve good generalization to new objects. The generated representation is suitable for object recognition. Performance degrades if the network is trained to learn multiple invariances simultaneously.

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42. Information Loss in an Optimal Maximum Likelihood Decoding

Inés Samengo

https://doi.org/10.1162/089976602317318947 | pp. 771-779

The mutual information between a set of stimuli and the elicited neural responses is compared to the corresponding decoded information. The decoding procedure is presented as an artificial distortion of the joint probabilities between stimuli and responses. The information loss is quantified. Whenever the probabilities are only slightly distorted, the information loss is shown to be quadratic in the distortion.

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43. The Reliability of the Stochastic Active Rotator

Khashayar Pakdaman

https://doi.org/10.1162/089976602317318956 | pp. 781-792

The reliability of firing of excitable-oscillating systems is studied through the response of the active rotator, a neuronal model evolving on the unit circle, to white gaussian noise. A stochastic return map is introduced that captures the behavior of the model. This map has two fixed points: one stable and the other unstable. Iterates of all initial conditions except the unstable point tend to the stable fixed point for almost all input realizations. This means that to a given input realization, there corresponds a unique asymptotic response. In this way, repetitive stimulation with the same segment of noise realization evokes, after possibly a transient time, the same response in the active rotator. In other words, this model responds reliably to such inputs. It is argued that this results from the nonuniform motion of the active rotator around the unit circle and that similar results hold for other neuronal models whose dynamics can be approximated by phase dynamics similar to the active rotator.

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44. A Proposed Function for Hippocampal Theta Rhythm: Separate Phases of Encoding and Retrieval Enhance Reversal of Prior Learning

Michael E. Hasselmo, Clara Bodelón, Bradley P. Wyble

https://doi.org/10.1162/089976602317318965 | pp. 793-817

The theta rhythm appears in the rat hippocampal electroencephalogram during exploration and shows phase locking to stimulus acquisition. Lesions that block theta rhythm impair performance in tasks requiring reversal of prior learning, including reversal in a T-maze, where associations between one arm location and

food reward need to be extinguished in favor of associations between the opposite arm location and food reward. Here, a hippocampal model shows how theta rhythm could be important for reversal in this task by providing separate functional phases during each 100-300 msec cycle, consistent with physiological data. In the model, effective encoding of new associations occurs in the phase when synaptic input from entorhinal cortex is strong and long-term potentiation (LTP) of excitatory connections arising from hippocampal region CA3 is strong, but synaptic currents arising from region CA3 input are weak (to prevent interference from prior learned associations). Retrieval of old associations occurs in the phase when entorhinal input is weak and synaptic input from region CA3 is strong, but when depotentiation occurs at synapses from CA3 (to allow extinction of prior learned associations that do not match current input). These phasic changes require that LTP at synapses arising from region CA3 should be strongest at the phase when synaptic transmission at these synapses is weakest. Consistent with these requirements, our recent data show that synaptic transmission in stratum radiatum is weakest at the positive peak of local theta, which is when previous data show that induction of LTP is strongest in this layer.

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45. Self-Organization in the Basal Ganglia with Modulation of Reinforcement Signals

Hiroyuki Nakahara, ■■■-■■■ ■■■■■, Okihide Hikosaka

https://doi.org/10.1162/089976602317318974 | pp. 819-844

Self-organization is one of fundamental brain computations for forming efficient representations of information. Experimental support for this idea has been largely limited to the developmental and reorganizational formation of neural circuits in the sensory cortices. We now propose that self-organization may also play an important role in short-term synaptic changes in reward-driven voluntary behaviors. It has recently been shown that many neurons in the basal ganglia change their sensory responses flexibly in relation to rewards. Our computational model proposes that the rapid changes in striatal projection neurons depend on the subtle balance between the Hebb-type mechanisms of excitation and inhibition, which are modulated by reinforcement signals. Simulations based on the model are shown to produce various types of neural activity similar to those found in experiments.

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46. Population Computation of Vectorial Transformations

Pierre Baraduc, Emmanuel Guigon

https://doi.org/10.1162/089976602317318983 | pp. 845-871

Many neurons of the central nervous system are broadly tuned to some sensory or motor variables. This property allows one to assign to each neuron a preferred attribute (PA). The width of tuning curves and the distribution of PAs in a population of neurons tuned to a given variable define the collective behavior of the population. In this article, we study the relationship of the nature of the tuning curves, the distribution of PAs, and computational properties of linear neuronal populations. We show that noise-resistant distributed linear algebraic processing and learning can be implemented by a population of cosine tuned neurons assuming a nonuniform but regular distribution of PAs. We extend these results analytically to the noncosine tuning and uniform distribution case and show with a numerical simulation that the results remain valid for a nonuniform regular distribution of PAs for broad noncosine tuning curves. These observations provide a theoretical basis for modeling general nonlinear sensorimotor transformations as sets of local linearized

representations.

47. Redistribution of Synaptic Efficacy Supports Stable Pattern Learning in Neural Networks

Gail A. Carpenter, Boriana L. Milenova

https://doi.org/10.1162/089976602317318992 | pp. 873-888

Markram and Tsodyks, by showing that the elevated synaptic efficacy observed with single-pulse long-term potentiation (LTP) measurements disappears with higher-frequency test pulses, have critically challenged the conventional assumption that LTP reflects a general gain increase. This observed change in frequency dependence during synaptic potentiation is called redistribution of synaptic efficacy (RSE). RSE is here seen as the local realization of a global design principle in a neural network for pattern coding. The underlying computational model posits an adaptive threshold rather than a multiplicative weight as the elementary unit of long-term memory. A distributed instar learning law allows thresholds to increase only monotonically, but adaptation has a bidirectional effect on the model postsynaptic potential. At each synapse, threshold increases implement pattern selectivity via a frequency-dependent signal component, while a complementary frequency-independent component nonspecifically strengthens the path. This synaptic balance produces changes in frequency dependence that are robustly similar to those observed by Markram and Tsodyks. The network design therefore suggests a functional purpose for RSE, which, by helping to bound total memory change, supports a distributed coding scheme that is stable with fast as well as slow learning. Multiplicative weights have served as a cornerstone for models of physiological data and neural systems for decades. Although the model discussed here does not implement detailed physiology of synaptic transmission, its new learning laws operate in a network architecture that suggests how recently discovered synaptic computations such as RSE may help produce new network capabilities such as learning that is fast, stable, and distributed.

48. Mean-Field Approaches to Independent Component Analysis

Pedro Højen-Sørensen, Ole Winther, Lars Kai Hansen

https://doi.org/10.1162/089976602317319009 | pp. 889-918

We develop mean-field approaches for probabilistic independent component analysis (ICA). The sources are estimated from the mean of their posterior distribution and the mixing matrix (and noise level) is estimated by maximum a posteriori (MAP). The latter requires the computation of (a good approximation to) the correlations between sources. For this purpose, we investigate three increasingly advanced mean-field methods: the variational (also known as naive mean field) approach, linear response corrections, and an adaptive version of the Thouless, Anderson and Palmer (1977) (TAP) mean-field approach, which is due to Opper and Winther (2001). The resulting algorithms are tested on a number of problems. On synthetic data, the advanced mean-field approaches are able to recover the correct mixing matrix in cases where the variational meanfield theory fails. For handwritten digits, sparse encoding is achieved using nonnegative source and mixing priors. For speech, the mean-field method is able to separate in the underdetermined (overcomplete) case of two sensors and three sources. One major advantage of the proposed method is its generality and algorithmic simplicity. Finally, we point out several possible extensions of the approaches developed here.

49. Neural Networks with Local Receptive Fields and Superlinear VC Dimension

Michael Schmitt

https://doi.org/10.1162/089976602317319018 | pp. 919-956

Local receptive field neurons comprise such well-known and widely used unit types as radial basis function (RBF) neurons and neurons with center-surround receptive field. We study the Vapnik-Chervonenkis (VC) dimension of feedforward neural networks with one hidden layer of these units. For several variants of local receptive field neurons, we show that the VC dimension of these networks is superlinear. In particular, we establish the bound Omega(W log k) for any reasonably sized network with W parameters and k hidden nodes. This bound is shown to hold for discrete center-surround receptive field neurons, which are physiologically relevant models of cells in the mammalian visual system, for neurons computing a difference of gaussians, which are popular in computational vision, and for standard RBF neurons, a major alternative to sigmoidal neurons in artificial neural networks. The result for RBF neural networks is of particular interest since it answers a question that has been open for several years. The results also give rise to lower bounds for networks with fixed input dimension. Regarding constants, all bounds are larger than those known thus far for similar architectures with sigmoidal neurons. The superlinear lower bounds contrast with linear upper bounds for single local receptive field neurons also derived here.

50. A Population Study of Integrate-and-Fire-or-Burst Neurons

A. Casti, Ahmet Omurtag, Andrew Sornborger, E. Kaplan, B. W. Knight, Jonathan D. Victor, L. Sirovich

https://doi.org/10.1162/089976602753633349 | pp. 957-986

Any realistic model of the neuronal pathway from the retina to the visual cortex (V1) must account for the bursting behavior of neurons in the lateral geniculate nucleus (LGN). A robust but minimal model, the integrate-and-fire-or-burst (IFB) model, has recently been proposed for individual LGN neurons. Based on this, we derive a dynamic population model and study a population of such LGN cells. This population model, the first simulation of its kind evolving in a two-dimensional phase space, is used to study the behavior of bursting populations in response to diverse stimulus conditions.

51. Stable Propagation of Activity Pulses in Populations of Spiking Neurons

Werner M. Kistler, Wulfram Gerstner

https://doi.org/10.1162/089976602753633358 | pp. 987-997

We investigate the propagation of pulses of spike activity in a neuronal network with feedforward couplings. The neurons are of the spike-response type with a firing probability that depends linearly on the membrane potential. After firing, neurons enter a phase of refractoriness. Spike packets are described in terms of the moments of the firing-time distribution so as to allow for an analytical treatment of the evolution of the spike packet as it propagates from one layer to the next. Analytical results and simulations show that depending on the synaptic coupling strength, a stable propagation of the packet with constant waveform is possible. Crucial for this observation is neither the existence of a firing threshold nor a sigmoidal gain function--both are absent in our model--but the refractory behavior of the neurons.

52. Population Coding and Decoding in a Neural Field: A Computational Study

Si Wu, ■■■-■■■ ■■■■■, Hiroyuki Nakahara

https://doi.org/10.1162/089976602753633367 | pp. 999-1026

This study uses a neural field model to investigate computational aspects of population coding and decoding when the stimulus is a single variable. A general prototype model for the encoding process is proposed, in which neural responses are correlated, with strength specified by a gaussian function of their difference in preferred stimuli. Based on the model, we study the effect of correlation on the Fisher information, compare the performances of three decoding methods that differ in the amount of encoding information being used, and investigate the implementation of the three methods by using a recurrent network. This study not only rediscovers main results in existing literatures in a unified way, but also reveals important new features, especially when the neural correlation is strong. As the neural correlation of firing becomes larger, the Fisher information decreases drastically. We confirm that as the width of correlation increases, the Fisher information saturates and no longer increases in proportion to the number of neurons. However, we prove that as the width increases further--wider than (sqrt)2 times the effective width of the turning function--the Fisher information increases again, and it increases without limit in proportion to the number of neurons. Furthermore, we clarify the asymptotic efficiency of the maximum likelihood inference (MLI) type of decoding methods for correlated neural signals. It shows that when the correlation covers a nonlocal range of population (excepting the uniform correlation and when the noise is extremely small), the MLI type of method, whose decoding error satisfies the Cauchy-type distribution, is not asymptotically efficient. This implies that the variance is no longer adequate to measure decoding accuracy.

53. The Influence of Limit Cycle Topology on the Phase Resetting Curve

Sorinel A. Oprisan, Carmen C. Canavier

https://doi.org/10.1162/089976602753633376 | pp. 1027-1057

Understanding the phenomenology of phase resetting is an essential step toward developing a formalism for the analysis of circuits composed of bursting neurons that receive multiple, and sometimes overlapping, inputs. If we are to use phase-resetting methods to analyze these circuits, we can either generate phase-resetting curves (PRCs) for all possible inputs and combinations of inputs, or we can develop an understanding of how to construct PRCs for arbitrary perturbations of a given neuron. The latter strategy is the goal of this study. We present a geometrical derivation of phase resetting of neural limit cycle oscillators in response to short current pulses. A geometrical phase is defined as the distance traveled along the limit cycle in the appropriate phase space. The perturbations in current are treated as displacements in the direction corresponding to membrane voltage. We show that for type I oscillators, the direction of a perturbation in current is nearly tangent to the limit cycle; hence, the projection of the displacement in voltage onto the limit cycle is sufficient to give the geometrical phase resetting. In order to obtain the phase resetting in terms of elapsed time or temporal phase, a mapping between geometrical and temporal phase is obtained empirically and used to make the conversion. This mapping is shown to be an invariant of the dynamics. Perturbations in current applied to type II oscillators produce significant normal displacements from the limit cycle, so the difference in angular velocity at displaced points compared to the angular velocity on the limit cycle must be taken into account. Empirical attempts to correct for differences in angular velocity

(amplitude versus phase effects in terms of a circular coordinate system) during relaxation back to the limit cycle achieved some success in the construction of phase-resetting curves for type II model oscillators. The ultimate goal of this work is the extension of these techniques to biological circuits comprising type II neural oscillators, which appear frequently in identified central pattern-generating circuits.

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54. Fields as Limit Functions of Stochastic Discrimination and Their Adaptability

Philip Van Loocke

https://doi.org/10.1162/089976602753633385 | pp. 1059-1070

For a particular type of elementary function, stochastic discrimination is shown to have an analytic limit function. Classifications can be performed directly by this limit function instead of by a sampling procedure. The limit function has an interpretation in terms of fields that originate from the training examples of a classification problem. Fields depend on the global configuration of the training points. The classification of a point in input space is known when the contributions of all fields are summed. Two modifications of the limit function are proposed. First, for nonlinear problems like high-dimensional parity problems, fields can be quantized. This leads to classification functions with perfect generalization for high-dimensional parity problems. Second, fields can be provided with adaptable amplitudes. The classification corresponding to a limit function is taken as an initialization; subsequently, amplitudes are adapted until an error function for the test set reaches minimal value. It is illustrated that this increases the performance of stochastic discrimination. Due to the nature of the fields, generalization improves even if the amplitude of every training example is adaptable.

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55. Learning to Recognize Three-Dimensional Objects

Dan Roth, Ming–Hsuan Yang, Narendra Ahuja

https://doi.org/10.1162/089976602753633394 | pp. 1071-1103

A learning account for the problem of object recognition is developed within the probably approximately correct (PAC) model of learnability. The key assumption underlying this work is that objects can be recognized (or discriminated) using simple representations in terms of syntactically simple relations over the raw image. Although the potential number of these simple relations could be huge, only a few of them are actually present in each observed image, and a fairly small number of those observed are relevant to discriminating an object. We show that these properties can be exploited to yield an efficient learning approach in terms of sample and computational complexity within the PAC model. No assumptions are needed on the distribution of the observed objects, and the learning performance is quantified relative to its experience. Most important, the success of learning an object representation is naturally tied to the ability to represent it as a function of some intermediate representations extracted from the image. We evaluate this approach in a large-scale experimental study in which the SNoW learning architecture is used to learn representations for the 100 objects in the Columbia Object Image Library. Experimental results exhibit good generalization and robustness properties of the SNoW-based method relative to other approaches. SNoW's recognition rate degrades more gracefully when the training data contains fewer views, and it shows similar behavior in some preliminary experiments with partially occluded objects.

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56. A Parallel Mixture of SVMs for Very Large Scale Problems

Ronan Collobert, Samy Bengio, Yoshua Bengio

https://doi.org/10.1162/089976602753633402 | pp. 1105-1114

Support vector machines (SVMs) are the state-of-the-art models for many classification problems, but they suffer from the complexity of their training algorithm, which is at least quadratic with respect to the number of examples. Hence, it is hopeless to try to solve real-life problems having more than a few hundred thousand examples with SVMs. This article proposes a new mixture of SVMs that can be easily implemented in parallel and where each SVM is trained on a small subset of the whole data set. Experiments on a large benchmark data set (Forest) yielded significant time improvement (time complexity appears empirically to locally grow linearly with the number of examples). In addition, and surprisingly, a significant improvement in generalization was observed.

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57. Bayesian Framework for Least-Squares Support Vector Machine Classifiers, Gaussian Processes, and Kernel Fisher Discriminant Analysis

Tony Van Gestel, Johan A. K. Suykens, Gert Lanckriet, Adriaan Lambrechts, Bart De Moor, Joos Vandewalle

https://doi.org/10.1162/089976602753633411 | pp. 1115-1147

The Bayesian evidence framework has been successfully applied to the design of multilayer perceptrons (MLPs) in the work of MacKay. Nevertheless, the training of MLPs suffers from drawbacks like the nonconvex optimization problem and the choice of the number of hidden units. In support vector machines (SVMs) for classification, as introduced by Vapnik, a nonlinear decision boundary is obtained by mapping the input vector first in a nonlinear way to a high-dimensional kernel-induced feature space in which a linear large margin classifier is constructed. Practical expressions are formulated in the dual space in terms of the related kernel function, and the solution follows from a (convex) quadratic programming (QP) problem. In least-squares SVMs (LS-SVMs), the SVM problem formulation is modified by introducing a least-squares cost function and equality instead of inequality constraints, and the solution follows from a linear system in the dual space. Implicitly, the least-squares formulation corresponds to a regression formulation and is also related to kernel Fisher discriminant analysis. The least-squares regression formulation has advantages for deriving analytic expressions in a Bayesian evidence framework, in contrast to the classification formulations used, for example, in gaussian processes (GPs). The LS-SVM formulation has clear primal-dual interpretations, and without the bias term, one explicitly constructs a model that yields the same expressions as have been obtained with GPs for regression. In this article, the Bayesian evidence framework is combined with the LS-SVM classifier formulation. Starting from the feature space formulation, analytic expressions are obtained in the dual space on the different levels of Bayesian inference, while posterior class probabilities are obtained by marginalizing over the model parameters. Empirical results obtained on 10 public domain data sets show that the LS-SVM classifier designed within the Bayesian evidence framework consistently yields good generalization performances.

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58. Neural Network Pruning with Tukey-Kramer Multiple Comparison Procedure

Donald E. Duckro, Dennis W. Quinn, Samuel J. Gardner

https://doi.org/10.1162/089976602753633420 | pp. 1149-1168

Reducing a neural network's complexity improves the ability of the network to generalize future examples. Like an overfitted regression function, neural networks may miss their target because of the excessive degrees of freedom stored up in unnecessary parameters. Over the past decade, the subject of pruning networks produced nonstatistical algorithms like Skeletonization, Optimal Brain Damage, and Optimal Brain Surgeon as methods to remove connections with the least salience. The method proposed here uses the bootstrap algorithm to estimate the distribution of the model parameter saliences. Statistical multiple comparison procedures are then used to make pruning decisions. We show this method compares well with Optimal Brain Surgeon in terms of ability to prune and the resulting network performance.

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59. Products of Gaussians and Probabilistic Minor Component Analysis

Christopher K. I. Williams, Felix Agakov

https://doi.org/10.1162/089976602753633439 | pp. 1169-1182

Recently, Hinton introduced the products of experts architecture for density estimation, where individual expert probabilities are multiplied and renormalized. We consider products of gaussian "pancakes" equally elongated in all directions except one and prove that the maximum likelihood solution for the model gives rise to a minor component analysis solution. We also discuss the covariance structure of sums and products of gaussian pancakes or one-factor probabilistic principal component analysis models.

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60. Optimization of the Kernel Functions in a Probabilistic Neural Network Analyzing the Local Pattern Distribution

Ingo Galleske, Juan Castellanos

https://doi.org/10.1162/089976602753633448 | pp. 1183-1194

This article proposes a procedure for the automatic determination of the elements of the covariance matrix of the gaussian kernel function of probabilistic neural networks. Two matrices, a rotation matrix and a matrix of variances, can be calculated by analyzing the local environment of each training pattern. The combination of them will form the covariance matrix of each training pattern. This automation has two advantages: First, it will free the neural network designer from indicating the complete covariance matrix, and second, it will result in a network with better generalization ability than the original model. A variation of the famous two-spiral problem and real-world examples from the UCI Machine Learning Repository will show a classification rate not only better than the original probabilistic neural network but also that this model can outperform other well-known classification techniques.

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61. Methods for Binary Multidimensional Scaling

Douglas L. T. Rohde

https://doi.org/10.1162/089976602753633457 | pp. 1195-1232

Multidimensional scaling (MDS) is the process of transforming a set of points in a high-dimensional space to a lower-dimensional one while preserving the relative distances between pairs of points. Although effective methods have been developed for solving a variety of MDS problems, they mainly depend on the vectors in

the lower-dimensional space having real-valued components. For some applications, the training of neural networks in particular, it is preferable or necessary to obtain vectors in a discrete, binary space. Unfortunately, MDS into a low-dimensional discrete space appears to be a significantly harder problem than MDS into a continuous space. This article introduces and analyzes several methods for performing approximately optimized binary MDS.

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62. Cosine Tuning Minimizes Motor Errors

Emanuel Todorov

https://doi.org/10.1162/089976602753712918 | pp. 1233-1260

Cosine tuning is ubiquitous in the motor system, yet a satisfying explanation of its origin is lacking. Here we argue that cosine tuning minimizes expected errors in force production, which makes it a natural choice for activating muscles and neurons in the final stages of motor processing. Our results are based on the empirically observed scaling of neuromotor noise, whose standard deviation is a linear function of the mean. Such scaling predicts a reduction of net force errors when redundant actuators pull in the same direction. We confirm this prediction by comparing forces produced with one versus two hands and generalize it across directions. Under the resulting neuromotor noise model, we prove that the optimal activation profile is a (possibly truncated) cosine--for arbitrary dimensionality of the workspace, distribution of force directions, correlated or uncorrelated noise, with or without a separate cocontraction command. The model predicts a negative force bias, truncated cosine tuning at low muscle cocontraction levels, and misalignment of preferred directions and lines of action for nonuniform muscle distributions. All predictions are supported by experimental data.

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63. SMEM Algorithm Is Not Fully Compatible with Maximum-Likelihood Framework

Akihiro Minagawa, Norio Tagawa, Toshiyuki Tanaka

https://doi.org/10.1162/089976602753712927 | pp. 1261-1266

The expectation-maximization (EM) algorithm with split-and-merge operations (SMEM algorithm) proposed by Ueda, Nakano, Ghahramani, and Hinton (2000) is a nonlocal searching method, applicable to mixture models, for relaxing the local optimum property of the EM algorithm. In this article, we point out that the SMEM algorithm uses the acceptance-rejection evaluation method, which may pick up a distribution with smaller likelihood, and demonstrate that an increase in likelihood can then be guaranteed only by comparing log likelihoods.

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64. A Note on the Decomposition Methods for Support Vector Regression

Shuo-Peng Liao, Hsuan-Tien Lin, Chih■Jen Lin

https://doi.org/10.1162/089976602753712936 | pp. 1267-1281

The dual formulation of support vector regression involves two closely related sets of variables. When the decomposition method is used, many existing approaches use pairs of indices from these two sets as the working set. Basically, they select a base set first and then expand it so all indices are pairs. This makes the

implementation different from that for support vector classification. In addition, a larger optimization subproblem has to be solved in each iteration. We provide theoretical proofs and conduct experiments to show that using the base set as the working set leads to similar convergence (number of iterations). Therefore, by using a smaller working set while keeping a similar number of iterations, the program can be simpler and more efficient.

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65. An Image Analysis Algorithm for Dendritic Spines

Ingrid Y.Y. Koh, W. Brent Lindquist, Karen Zito, Esther A. Nimchinsky, Karel Svoboda

https://doi.org/10.1162/089976602753712945 | pp. 1283-1310

The structure of neuronal dendrites and their spines underlie the connectivity of neural networks. Dendrites, spines, and their dynamics are shaped by genetic programs as well as sensory experience. Dendritic structures and dynamics may therefore be important predictors of the function of neural networks. Based on new imaging approaches and increases in the speed of computation, it has become possible to acquire large sets of high-resolution optical micrographs of neuron structure at length scales small enough to resolve spines. This advance in data acquisition has not been accompanied by comparable advances in data analysis techniques; the analysis of dendritic and spine morphology is still accomplished largely manually. In addition to being extremely time intensive, manual analysis also introduces systematic and hard-to-characterize biases. We present a geometric approach for automatically detecting and quantifying the three-dimensional structure of dendritic spines from stacks of image data acquired using laser scanning microscopy. We present results on the measurement of dendritic spine length, volume, density, and shape classification for both static and time-lapse images of dendrites of hippocampal pyramidal neurons. For spine length and density, the automated measurements in static images are compared with manual measurements. Comparisons are also made between automated and manual spine length measurements for a time-series data set. The algorithm performs well compared to a human analyzer, especially on time-series data. Automated analysis of dendritic spine morphology will enable objective analysis of large morphological data sets. The approaches presented here are generalizable to other aspects of neuronal morphology.

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66. Multiplicative Synaptic Normalization and a Nonlinear Hebb Rule Underlie a Neurotrophic Model of Competitive Synaptic Plasticity

Terry Elliott, N. R. Shadbolt

https://doi.org/10.1162/089976602753712954 | pp. 1311-1322

Synaptic normalization is used to enforce competitive dynamics in many models of developmental synaptic plasticity. In linear and semilinear Hebbian models, multiplicative synaptic normalization fails to segregate afferents whose activity patterns are positively correlated. To achieve this, the biologically problematic device of subtractive synaptic normalization must be used instead. Our own model of competition for neurotrophic support, which can segregate positively correlated afferents, was developed in part in an attempt to overcome these problems by removing the need for synaptic normalization altogether. However, we now show that the dynamics of our model decompose into two decoupled subspaces, with competitive dynamics being implemented in one of them through a nonlinear Hebb rule and multiplicative synaptic normalization. This normalization is "emergent" rather than imposed. We argue that these observations

permit biologically plausible forms of synaptic normalization to be viewed as abstract and general descriptions of the underlying biology in certain scaleless models of synaptic plasticity.

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67. Energy-Efficient Coding with Discrete Stochastic Events

Susanne Schreiber, Christian K. Machens, Andreas V. M. Herz, Simon B. Laughlin
https://doi.org/10.1162/089976602753712963 | pp. 1323-1346

We investigate the energy efficiency of signaling mechanisms that transfer information by means of discrete stochastic events, such as the opening or closing of an ion channel. Using a simple model for the generation of graded electrical signals by sodium and potassium channels, we find optimum numbers of channels that maximize energy efficiency. The optima depend on several factors: the relative magnitudes of the signaling cost (current flow through channels), the fixed cost of maintaining the system, the reliability of the input, additional sources of noise, and the relative costs of upstream and downstream mechanisms. We also analyze how the statistics of input signals influence energy efficiency. We find that energy-efficient signal ensembles favor a bimodal distribution of channel activations and contain only a very small fraction of large inputs when energy is scarce. We conclude that when energy use is a significant constraint, trade-offs between information transfer and energy can strongly influence the number of signaling molecules and synapses used by neurons and the manner in which these mechanisms represent information.

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68. Multiple Model-Based Reinforcement Learning

Kenji Doya, Kazuyuki Samejima, Ken-ichi Katagiri, Mitsuo Kawato
https://doi.org/10.1162/089976602753712972 | pp. 1347-1369

We propose a modular reinforcement learning architecture for nonlinear, nonstationary control tasks, which we call multiple model-based reinforcement learning (MMRL). The basic idea is to decompose a complex task into multiple domains in space and time based on the predictability of the environmental dynamics. The system is composed of multiple modules, each of which consists of a state prediction model and a reinforcement learning controller. The "responsibility signal," which is given by the softmax function of the prediction errors, is used to weight the outputs of multiple modules, as well as to gate the learning of the prediction models and the reinforcement learning controllers. We formulate MMRL for both discrete-time, finite-state case and continuous-time, continuous-state case. The performance of MMRL was demonstrated for discrete case in a nonstationary hunting task in a grid world and for continuous case in a nonlinear, nonstationary control task of swinging up a pendulum with variable physical parameters.

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69. A Bayesian Approach to the Stereo Correspondence Problem

Jenny C. A. Read
https://doi.org/10.1162/089976602753712981 | pp. 1371-1392

I present a probabilistic approach to the stereo correspondence problem. Rather than trying to find a single solution in which each point in the left retina is assigned a partner in the right retina, all possible matches are considered simultaneously and assigned a probability of being correct. This approach is particularly suitable for stimuli where it is inappropriate to seek a unique partner for each retinal position--for instance, where

objects occlude each other, as in Panum's limiting case. The probability assigned to each match is based on a Bayesian analysis previously developed to explain psychophysical data (Read, 2002). This provides a convenient way to incorporate constraints that enable the ill-posed correspondence problem to be solved. The resulting model behaves plausibly for a variety of different stimuli.

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70. Learning Curves for Gaussian Process Regression: Approximations and Bounds

Peter Sollich, Anason Halees

https://doi.org/10.1162/089976602753712990 | pp. 1393-1428

We consider the problem of calculating learning curves (i.e., average generalization performance) of gaussian processes used for regression. On the basis of a simple expression for the generalization error, in terms of the eigenvalue decomposition of the covariance function, we derive a number of approximation schemes. We identify where these become exact and compare with existing bounds on learning curves; the new approximations, which can be used for any input space dimension, generally get substantially closer to the truth. We also study possible improvements to our approximations. Finally, we use a simple exactly solvable learning scenario to show that there are limits of principle on the quality of approximations and bounds expressible solely in terms of the eigenvalue spectrum of the covariance function.

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71. A Global Optimum Approach for One-Layer Neural Networks

Enrique Castillo, Óscar Fontenla-Romero, Bertha Guijarro■Berdiñas, Amparo Alonso■Betanzos

https://doi.org/10.1162/089976602753713007 | pp. 1429-1449

The article presents a method for learning the weights in one-layer feedforward neural networks minimizing either the sum of squared errors or the maximum absolute error, measured in the input scale. This leads to the existence of a global optimum that can be easily obtained solving linear systems of equations or linear programming problems, using much less computational power than the one associated with the standard methods. Another version of the method allows computing a large set of estimates for the weights, providing robust, mean or median, estimates for them, and the associated standard errors, which give a good measure for the quality of the fit. Later, the standard one-layer neural network algorithms are improved by learning the neural functions instead of assuming them known. A set of examples of applications is used to illustrate the methods. Finally, a comparison with other high-performance learning algorithms shows that the proposed methods are at least 10 times faster than the fastest standard algorithm used in the comparison.

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72. MLP in Layer-Wise Form with Applications to Weight Decay

Tommi Kärkkäinen

https://doi.org/10.1162/089976602753713016 | pp. 1451-1480

A simple and general calculus for the sensitivity analysis of a feedforward MLP network in a layer-wise form is presented. Based on the local optimality conditions, some consequences for the least-means-squares learning problem are stated and further discussed. Numerical experiments with formulation and comparison of different weight decay techniques are included.

73. Local Overfitting Control via Leverages

Gaétan Monari, Gérard Dreyfus

https://doi.org/10.1162/089976602753713025 | pp. 1481-1506

We present a novel approach to dealing with overfitting in black box models. It is based on the leverages of the samples, that is, on the influence that each observation has on the parameters of the model. Since overfitting is the consequence of the model specializing on specific data points during training, we present a selection method for nonlinear models based on the estimation of leverages and confidence intervals. It allows both the selection among various models of equivalent complexities corresponding to different minima of the cost function (e.g., neural nets with the same number of hidden units) and the selection among models having different complexities (e.g., neural nets with different numbers of hidden units). A complete model selection methodology is derived.

74. A Monte Carlo EM Approach for Partially Observable Diffusion Processes: Theory and Applications to Neural Networks

Javier R. Movellan, Paul Mineiro, Ruth Williams

https://doi.org/10.1162/08997660260028593 | pp. 1507-1544

We present a Monte Carlo approach for training partially observable diffusion processes. We apply the approach to diffusion networks, a stochastic version of continuous recurrent neural networks. The approach is aimed at learning probability distributions of continuous paths, not just expected values. Interestingly, the relevant activation statistics used by the learning rule presented here are inner products in the Hilbert space of square integrable functions. These inner products can be computed using Hebbian operations and do not require backpropagation of error signals. Moreover, standard kernel methods could potentially be applied to compute such inner products. We propose that the main reason that recurrent neural networks have not worked well in engineering applications (e.g., speech recognition) is that they implicitly rely on a very simplistic likelihood model. The diffusion network approach proposed here is much richer and may open new avenues for applications of recurrent neural networks. We present some analysis and simulations to support this view. Very encouraging results were obtained on a visual speech recognition task in which neural networks outperformed hidden Markov models.

75. Are Visual Cortex Maps Optimized for Coverage?

Miguel Á. Carreira-Perpiñán, Geoffrey J. Goodhill

https://doi.org/10.1162/08997660260028601 | pp. 1545-1560

The elegant regularity of maps of variables such as ocular dominance, orientation, and spatial frequency in primary visual cortex has prompted many people to suggest that their structure could be explained by an optimization principle. Up to now, the standard way to test this hypothesis has been to generate artificial maps by optimizing a hypothesized objective function and then to compare these artificial maps with real maps using a variety of quantitative criteria. If the artificial maps are similar to the real maps, this provides some evidence that the real cortex may be optimizing a similar function to the one hypothesized. Recently, a more direct method has been proposed for testing whether real maps represent local optima of an objective

function (Swindale, Shoham, Grinvald, Bonhoeffer, & Hübener, 2000). In this approach, the value of the hypothesized function is calculated for a real map, and then the real map is perturbed in certain ways and the function recalculated. If each of these perturbations leads to a worsening of the function, it is tempting to conclude that the real map is quite likely to represent a local optimum of that function. In this article, we argue that such perturbation results provide only weak evidence in favor of the optimization hypothesis.

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76. Kernel-Based Topographic Map Formation by Local Density Modeling

Marc M. Van Hulle

https://doi.org/10.1162/08997660260028610 | pp. 1561-1573

We introduce a new learning algorithm for kernel-based topographic map formation. The algorithm generates a gaussian mixture density model by individually adapting the gaussian kernels' centers and radii to the assumed gaussian local input densities.

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77. A Simple Model of Long-Term Spike Train Regularization

Relly Brandman, Mark Nelson

https://doi.org/10.1162/08997660260028629 | pp. 1575-1597

A simple model of spike generation is described that gives rise to negative correlations in the interspike interval (ISI) sequence and leads to long-term spike train regularization. This regularization can be seen by examining the variance of the kth-order interval distribution for large k (the times between spike i and spike i + k). The variance is much smaller than would be expected if successive ISIs were uncorrelated. Such regularizing effects have been observed in the spike trains of electrosensory afferent nerve fibers and can lead to dramatic improvement in the detectability of weak signals encoded in the spike train data (Ratnam & Nelson, 2000). Here, we present a simple neural model in which negative ISI correlations and long-term spike train regularization arise from refractory effects associated with a dynamic spike threshold. Our model is derived from a more detailed model of electrosensory afferent dynamics developed recently by other investigators (Chacron, Longtin, St.-Hilaire, & Maler, 2000;Chacron, Longtin, & Maler, 2001). The core of this model is a dynamic spike threshold that is transiently elevated following a spike and subsequently decays until the next spike is generated. Here, we present a simplified version-the linear adaptive threshold model-that contains a single state variable and three free parameters that control the mean and coefficient of variation of the spontaneous ISI distribution and the frequency characteristics of the driven response. We show that refractory effects associated with the dynamic threshold lead to regularization of the spike train on long timescales. Furthermore, we show that this regularization enhances the detectability of weak signals encoded by the linear adaptive threshold model. Although inspired by properties of electrosensory afferent nerve fibers, such regularizing effects may play an important role in other neural systems where weak signals must be reliably detected in noisy spike trains. When modeling a neuronal system that exhibits this type of ISI correlation structure, the linear adaptive threshold model may provide a more appropriate starting point than conventional renewal process models that lack long-term regularizing effects.

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78. Spatiotemporal Spike Encoding of a Continuous External Signal

Naoki Masuda, Kazuyuki Aihara
https://doi.org/10.1162/08997660260028638 | pp. 1599-1628

Interspike intervals of spikes emitted from an integrator neuron model of sensory neurons can encode input information represented as a continuous signal from a deterministic system. If a real brain uses spike timing as a means of information processing, other neurons receiving spatiotemporal spikes from such sensory neurons must also be capable of treating information included in deterministic interspike intervals. In this article, we examine functions of neurons modeling cortical neurons receiving spatiotemporal spikes from many sensory neurons. We show that such neuron models can encode stimulus information passed from the sensory model neurons in the form of interspike intervals. Each sensory neuron connected to the cortical neuron contributes equally to the information collection by the cortical neuron. Although the incident spike train to the cortical neuron is a superimposition of spike trains from many sensory neurons, it need not be decomposed into spike trains according to the input neurons. These results are also preserved for generalizations of sensory neurons such as a small amount of leak, noise, inhomogeneity in firing rates, or biases introduced in the phase distributions.

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79. Attractor Reliability Reveals Deterministic Structure in Neuronal Spike Trains

Paul Tiesinga, Jean■Marc Fellous, Terrence J. Sejnowski
https://doi.org/10.1162/08997660260028647 | pp. 1629-1650

When periodic current is injected into an integrate-and-fire model neuron, the voltage as a function of time converges from different initial conditions to an attractor that produces reproducible sequences of spikes. The attractor reliability is a measure of the stability of spike trains against intrinsic noise and is quantified here as the inverse of the number of distinct spike trains obtained in response to repeated presentations of the same stimulus. High reliability characterizes neurons that can support a spike-time code, unlike neurons with discharges forming a renewal process (such as a Poisson process). These two classes of responses cannot be distinguished using measures based on the spike-time histogram, but they can be identified by the attractor dynamics of spike trains, as shown here using a new method for calculating the attractor reliability. We applied these methods to spike trains obtained from current injection into cortical neurons recorded in vitro. These spike trains did not form a renewal process and had a higher reliability compared to renewal-like processes with the same spike-time histogram.

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80. Traveling Waves of Excitation in Neural Field Models: Equivalence of Rate Descriptions and Integrate-and-Fire Dynamics

Daniel Cremers, Andreas V. M. Herz
https://doi.org/10.1162/08997660260028656 | pp. 1651-1667

Field models provide an elegant mathematical framework to analyze large-scale patterns of neural activity. On the microscopic level, these models are usually based on either a firing-rate picture or integrate-and-fire dynamics. This article shows that in spite of the large conceptual differences between the two types of dynamics, both generate closely related plane-wave solutions. Furthermore, for a large group of models, estimates about the network connectivity derived from the speed of these plane waves only marginally

depend on the assumed class of microscopic dynamics. We derive quantitative results about this phenomenon and discuss consequences for the interpretation of experimental data.

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81. Attentional Recruitment of Inter-Areal Recurrent Networks for Selective Gain Control

Richard H. R. Hahnloser, Rodney J. Douglas, K. Hepp

https://doi.org/10.1162/08997660260028665 | pp. 1669-1689

There is strong anatomical and physiological evidence that neurons with large receptive fields located in higher visual areas are recurrently connected to neurons with smaller receptive fields in lower areas. We have previously described a minimal neuronal network architecture in which top-down attentional signals to large receptive field neurons can bias and selectively read out the bottom-up sensory information to small receptive field neurons (Hahnloser, Douglas, Mahowald, & Hepp, 1999). Here we study an enhanced model, where the role of attention is to recruit specific inter-areal feedback loops (e.g., drive neurons above firing threshold). We first illustrate the operation of recruitment on a simple example of visual stimulus selection. In the subsequent analysis, we find that attentional recruitment operates by dynamical modulation of signal amplification and response multistability. In particular, we find that attentional stimulus selection necessitates increased recruitment when the stimulus to be selected is of small contrast and of small distance away from distractor stimuli. The selectability of a low-contrast stimulus is dependent on the gain of attentional effects; for example, low-contrast stimuli can be selected only when attention enhances neural responses. However, the dependence of attentional selection on stimulus-distractor distance is not contingent on whether attention enhances or suppresses responses. The computational implications of attentional recruitment are that cortical circuits can behave as winner-take-all mechanisms of variable strength and can achieve close to optimal signal discrimination in the presence of external noise.

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82. CCCP Algorithms to Minimize the Bethe and Kikuchi Free Energies: Convergent Alternatives to Belief Propagation

Alan Yuille

https://doi.org/10.1162/08997660260028674 | pp. 1691-1722

This article introduces a class of discrete iterative algorithms that are provably convergent alternatives to belief propagation (BP) and generalized belief propagation (GBP). Our work builds on recent results by Yedidia, Freeman, and Weiss (2000), who showed that the fixed points of BP and GBP algorithms correspond to extrema of the Bethe and Kikuchi free energies, respectively. We obtain two algorithms by applying CCCP to the Bethe and Kikuchi free energies, respectively (CCCP is a procedure, introduced here, for obtaining discrete iterative algorithms by decomposing a cost function into a concave and a convex part). We implement our CCCP algorithms on two- and three-dimensional spin glasses and compare their results to BP and GBP. Our simulations show that the CCCP algorithms are stable and converge very quickly (the speed of CCCP is similar to that of BP and GBP). Unlike CCCP, BP will often not converge for these problems (GBP usually, but not always, converges). The results found by CCCP applied to the Bethe or Kikuchi free energies are equivalent, or slightly better than, those found by BP or GBP, respectively (when BP and GBP converge). Note that for these, and other problems, BP and GBP give very accurate results (see Yedidia et al., 2000), and failure to converge is their major error mode. Finally, we point out that our

algorithms have a large range of inference and learning applications.

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83. Fast Curvature Matrix-Vector Products for Second-Order Gradient Descent

Nicol N. Schraudolph

https://doi.org/10.1162/08997660260028683 | pp. 1723-1738

We propose a generic method for iteratively approximating various second-order gradient steps - Newton, Gauss-Newton, Levenberg-Marquardt, and natural gradient - in linear time per iteration, using special curvature matrix-vector products that can be computed in O(n). Two recent acceleration techniques for on-line learning, matrix momentum and stochastic meta-descent (SMD), implement this approach. Since both were originally derived by very different routes, this offers fresh insight into their operation, resulting in further improvements to SMD.

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84. Representation and Extrapolation in Multilayer Perceptrons

Antony Browne

https://doi.org/10.1162/08997660260028692 | pp. 1739-1754

To give an adequate explanation of cognition and perform certain practical tasks, connectionist systems must be able to extrapolate. This work explores the relationship between input representation and extrapolation, using simulations of multilayer perceptrons trained to model the identity function. It has been discovered that representation has a marked effect on extrapolation.

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85. Using Noise to Compute Error Surfaces in Connectionist Networks: A Novel Means of Reducing Catastrophic Forgetting

Robert M. French, Nick Chater

https://doi.org/10.1162/08997660260028700 | pp. 1755-1769

In error-driven distributed feedforward networks, new information typically interferes, sometimes severely, with previously learned information. We show how noise can be used to approximate the error surface of previously learned information. By combining this approximated error surface with the error surface associated with the new information to be learned, the network's retention of previously learned items can be improved and catastrophic interference significantly reduced. Further, we show that the noise-generated error surface is produced using only first-derivative information and without recourse to any explicit error information.

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86. Training Products of Experts by Minimizing Contrastive Divergence

Geoffrey E. Hinton

https://doi.org/10.1162/089976602760128018 | pp. 1771-1800

It is possible to combine multiple latent-variable models of the same data by multiplying their probability distributions together and then renormalizing. This way of combining individual "expert" models makes it hard to generate samples from the combined model but easy to infer the values of the latent variables of each expert, because the combination rule ensures that the latent variables of different experts are conditionally independent when given the data. A product of experts (PoE) is therefore an interesting candidate for a perceptual system in which rapid inference is vital and generation is unnecessary. Training a PoE by maximizing the likelihood of the data is difficult because it is hard even to approximate the derivatives of the renormalization term in the combination rule. Fortunately, a PoE can be trained using a different objective function called "contrastive divergence" whose derivatives with regard to the parameters can be approximated accurately and efficiently. Examples are presented of contrastive divergence learning using several types of expert on several types of data.

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87. Dynamic Approximation of Spatiotemporal Receptive Fields in Nonlinear Neural Field Models

Thomas Wennekers

https://doi.org/10.1162/089976602760128027 | pp. 1801-1825

This article presents an approximation method to reduce the spatiotemporal behavior of localized activation peaks (also called "bumps") in non-linear neural field equations to a set of coupled ordinary differential equations (ODEs) for only the amplitudes and tuning widths of these peaks. This enables a simplified analysis of steady-state receptive fields and their stability, as well as spatiotemporal point spread functions and dynamic tuning properties. A lowest-order approximation for peak amplitudes alone shows that much of the well-studied behavior of small neural systems (e.g., the Wilson-Cowan oscillator) should carry over to localized solutions in neural fields. Full spatiotemporal response profiles can further be reconstructed from this low-dimensional approximation. The method is applied to two standard neural field models: a one-layer model with difference-of-gaussians connectivity kernel and a two-layer excitatory-inhibitory network. Similar models have been previously employed in numerical studies addressing orientation tuning of cortical simple cells. Explicit formulas for tuning properties, instabilities, and oscillation frequencies are given, and exemplary spatiotemporal response functions, reconstructed from the low-dimensional approximation, are compared with full network simulations.

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88. Independent Components of Magnetoencephalography: Localization

Akaysha C. Tang, Barak A. Pearlmutter, Natalie A. Malaszenko, Dan Phung, Bethany C. Reeb

https://doi.org/10.1162/089976602760128036 | pp. 1827-1858

We applied second-order blind identification (SOBI), an independent component analysis method, to MEG data collected during cognitive tasks. We explored SOBI's ability to help isolate underlying neuronal sources with relatively poor signal-to-noise ratios, allowing their identification and localization. We compare localization of the SOBI-separated components to localization from unprocessed sensor signals, using an equivalent current dipole modeling method. For visual and somatosensory modalities, SOBI preprocessing resulted in components that can be localized to physiologically and anatomically meaningful locations. Furthermore, this preprocessing allowed the detection of neuronal source activations that were otherwise undetectable. This increased probability of neuronal source detection and localization can be particularly

beneficial for MEG studies of higher-level cognitive functions, which often have greater signal variability and degraded signal-to-noise ratios than sensory activation tasks.

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89. Robust Blind Source Separation by Beta Divergence

Mihoko Minami, Shinto Eguchi

https://doi.org/10.1162/089976602760128045 | pp. 1859-1886

Blind source separation is aimed at recovering original independent signals when their linear mixtures are observed. Various methods for estimating a recovering matrix have been proposed and applied to data in many fields, such as biological signal processing, communication engineering, and financial market data analysis. One problem these methods have is that they are often too sensitive to outliers, and the existence of a few outliers might change the estimate drastically. In this article, we propose a robust method of blind source separation based on the beta divergence. Shift parameters are explicitly included in our model instead of the conventional way which assumes that original signals have zero mean. The estimator gives smaller weights to possible outliers so that their influence on the estimate is weakened. Simulation results show that the proposed estimator significantly improves the performance over the existing methods when outliers exist; it keeps equal performance otherwise.

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90. Joint Entropy Maximization in Kernel-Based Topographic Maps

Marc M. Van Hulle

https://doi.org/10.1162/089976602760128054 | pp. 1887-1906

A new learning algorithm for kernel-based topographic map formation is introduced. The kernel parameters are adjusted individually so as to maximize the joint entropy of the kernel outputs. This is done by maximizing the differential entropies of the individual kernel outputs, given that the map's output redundancy, due to the kernel overlap, needs to be minimized. The latter is achieved by minimizing the mutual information between the kernel outputs. As a kernel, the (radial) incomplete gamma distribution is taken since, for a gaussian input density, the differential entropy of the kernel output will be maximal. Since the theoretically optimal joint entropy performance can be derived for the case of nonoverlapping gaussian mixture densities, a new clustering algorithm is suggested that uses this optimum as its "null" distribution. Finally, it is shown that the learning algorithm is similar to one that performs stochastic gradient descent on the Kullback-Leibler divergence for a heteroskedastic gaussian mixture density model.

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91. Sufficient Conditions for Error Backflow Convergence in Dynamical Recurrent Neural Networks

Alex Aussem

https://doi.org/10.1162/089976602760128063 | pp. 1907-1927

This article extends previous analysis of the gradient decay to a class of discrete-time fully recurrent networks, called dynamical recurrent neural networks, obtained by modeling synapses as finite impulse response (FIR) filters instead of multiplicative scalars. Using elementary matrix manipulations, we provide an upper bound on the norm of the weight matrix, ensuring that the gradient vector, when propagated in a

reverse manner in time through the error-propagation network, decays exponentially to zero. This bound applies to all recurrent FIR architecture proposals, as well as fixed-point recurrent networks, regardless of delay and connectivity. In addition, we show that the computational overhead of the learning algorithm can be reduced drastically by taking advantage of the exponential decay of the gradient.

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92. Bayesian A* Tree Search with Expected O(N) Node Expansions: Applications to Road Tracking

James M. Coughlan, Alan Yuille

https://doi.org/10.1162/089976602760128072 | pp. 1929-1958

Many perception, reasoning, and learning problems can be expressed as Bayesian inference. We point out that formulating a problem as Bayesian inference implies specifying a probability distribution on the ensemble of problem instances. This ensemble can be used for analyzing the expected complexity of algorithms and also the algorithm-independent limits of inference. We illustrate this problem by analyzing the complexity of tree search. In particular, we study the problem of road detection, as formulated by Geman and Jedynak (1996). We prove that the expected convergence is linear in the size of the road (the depth of the tree) even though the worst-case performance is exponential. We also put a bound on the constant of the convergence and place a bound on the error rates.

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93. Training v-Support Vector Regression: Theory and Algorithms

Chih-Chung Chang, Chih■Jen Lin

https://doi.org/10.1162/089976602760128081 | pp. 1959-1977

We discuss the relation between epsilon-support vector regression (epsilon-SVR) and nu-support vector regression (nu-SVR). In particular, we focus on properties that are different from those of C-support vector classification (C-SVC) and nu-support vector classification (nu-SVC). We then discuss some issues that do not occur in the case of classification: the possible range of epsilon and the scaling of target values. A practical decomposition method for nu-SVR is implemented, and computational experiments are conducted. We show some interesting numerical observations specific to regression.

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94. On the Problem in Model Selection of Neural Network Regression in Overrealizable Scenario

Katsuyuki Hagiwara

https://doi.org/10.1162/089976602760128090 | pp. 1979-2002

In considering a statistical model selection of neural networks and radial basis functions under an overrealizable case, the problem of unidentifiability emerges. Because the model selection criterion is an unbiased estimator of the generalization error based on the training error, this article analyzes the expected training error and the expected generalization error of neural networks and radial basis functions in overrealizable cases and clarifies the difference from regular models, for which identifiability holds. As a special case of an overrealizable scenario, we assumed a gaussian noise sequence as training data. In the least-squares estimation under this assumption, we first formulated the problem, in which the calculation of

the expected errors of unidentifiable networks is reduced to the calculation of the expectation of the supremum of the chi2 process. Under this formulation, we gave an upper bound of the expected training error and a lower bound of the expected generalization error, where the generalization is measured at a set of training inputs. Furthermore, we gave stochastic bounds on the training error and the generalization error. The obtained upper bound of the expected training error is smaller than in regular models, and the lower bound of the expected generalization error is larger than in regular models. The result tells us that the degree of overfitting in neural networks and radial basis functions is higher than in regular models. Correspondingly, it also tells us that the generalization capability is worse than in the case of regular models. The article may be enough to show a difference between neural networks and regular models in the context of the least-squares estimation in a simple situation. This is a first step in constructing a model selection criterion in an overrealizable case. Further important problems in this direction are also included in this article.

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95. Scalable Hybrid Computation with Spikes

Rahul Sarpeshkar, M. J. O'Halloran

https://doi.org/10.1162/089976602320263971 | pp. 2003-2038

We outline a hybrid analog-digital scheme for computing with three important features that enable it to scale to systems of large complexity: First, like digital computation, which uses several one-bit precise logical units to collectively compute a precise answer to a computation, the hybrid scheme uses several moderate-precision analog units to collectively compute a precise answer to a computation. Second, frequent discrete signal restoration of the analog information prevents analog noise and offset from degrading the computation. And, third, a state machine enables complex computations to be created using a sequence of elementary computations. A natural choice for implementing this hybrid scheme is one based on spikes because spike-count codes are digital, while spike-time codes are analog. We illustrate how spikes afford easy ways to implement all three components of scalable hybrid computation. First, as an important example of distributed analog computation, we show how spikes can create a distributed modular representation of an analog number by implementing digital carry interactions between spiking analog neurons. Second, we show how signal restoration may be performed by recursive spike-count quantization of spike-time codes. And, third, we use spikes from an analog dynamical system to trigger state transitions in a digital dynamical system, which reconfigures the analog dynamical system using a binary control vector; such feedback interactions between analog and digital dynamical systems create a hybrid state machine (HSM). The HSM extends and expands the concept of a digital finite-state-machine to the hybrid domain. We present experimental data from a two-neuron HSM on a chip that implements error-correcting analog-to-digital conversion with the concurrent use of spike-time and spike-count codes. We also present experimental data from silicon circuits that implement HSM-based pattern recognition using spike-time synchrony. We outline how HSMs may be used to perform learning, vector quantization, spike pattern recognition and generation, and how they may be reconfigured.

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96. Learning Nonregular Languages: A Comparison of Simple Recurrent Networks and LSTM

Jürgen Schmidhuber, Felix A. Gers, Douglas Eck

https://doi.org/10.1162/089976602320263980 | pp. 2039-2041

In response to Rodriguez's recent article (2001), we compare the performance of simple recurrent nets and long short-term memory recurrent nets on context-free and context-sensitive languages.

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97. Center-Crossing Recurrent Neural Networks for the Evolution of Rhythmic Behavior

Boonyanit Mathayomchan, Randall D. Beer

https://doi.org/10.1162/089976602320263999 | pp. 2043-2051

A center-crossing recurrent neural network is one in which the null-(hyper)surfaces of each neuron intersect at their exact centers of symmetry, ensuring that each neuron's activation function is centered over the range of net inputs that it receives. We demonstrate that relative to a random initial population, seeding the initial population of an evolutionary search with center-crossing networks significantly improves both the frequency and the speed with which high-fitness oscillatory circuits evolve on a simple walking task. The improvement is especially striking at low mutation variances. Our results suggest that seeding with center-crossing networks may often be beneficial, since a wider range of dynamics is more likely to be easily accessible from a population of center-crossing networks than from a population of random networks.

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98. Reply to Carreira-Perpiñán and Goodhill

Nicholas V. Swindale, Doron Shoham, Amiram Grinvald, Tobias Bonhoeffer, Mark Hübener

https://doi.org/10.1162/089976602320264006 | pp. 2053-2056

September 01 2002 Reply to Carreira-Perpiñán and Goodhill In Special Collection: CogNet Nicholas V. Swindale, Nicholas V. Swindale Department of Ophthalmology, University of British Columbia, Vancouver, B.C., V5Z 3N9, swindale@interchange.ubc.ca Search for other works by this author on: This Site Google Scholar Doron Shoham, Doron Shoham Weizmann Institute of Science, Department of Neurobiology, Rehovot 76100, Israel Search for other works by this author on: This Site Google Scholar Amiram Grinvald, Amiram Grinvald Weizmann Institute of Science, Department of Neurobiology, Rehovot 76100, Israel, Amiram.Grinvald@weizmann.ac.il Search for other works by this author on: This Site Google Scholar Tobias Bonhoeffer, Tobias Bonhoeffer Max-Planck-Institut für Neurobiologie, D-82152 Martinsried, Germany, tobias.bonhoeffer@neuro.mpg.de Search for other works by this author on: This Site Google Scholar Mark Hübener Mark Hübener Max-Planck-Institut für Neurobiologie, D-82152 Martinsried, Germany, mark@neuro.mpg.de Search for other works by this author on: This Site Google Scholar Author and Article Information Nicholas V. Swindale Department of Ophthalmology, University of British Columbia, Vancouver, B.C., V5Z 3N9, swindale@interchange.ubc.ca Doron Shoham Weizmann Institute of Science, Department of Neurobiology, Rehovot 76100, Israel Amiram Grinvald Weizmann Institute of Science, Department of Neurobiology, Rehovot 76100, Israel, Amiram.Grinvald@weizmann.ac.il Tobias Bonhoeffer Max-Planck-Institut für Neurobiologie, D-82152 Martinsried, Germany, tobias.bonhoeffer@neuro.mpg.de Mark Hübener Max-Planck-Institut für Neurobiologie, D-82152 Martinsried, Germany, mark@neuro.mpg.de Received: November 20 2001 Accepted: February 25 2002 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 2002 Massachusetts Institute of Technology2002 Neural Computation (2002) 14 (9): 2053–2056. https://doi.org/10.1162/089976602320264006 Article history Received: November 20 2001 Accepted:

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99. Dynamics of the Firing Probability of Noisy Integrate-and-Fire Neurons

Nicolas Fourcaud■Trocmé, Nicolas Brunel

https://doi.org/10.1162/089976602320264015 | pp. 2057-2110

Cortical neurons in vivo undergo a continuous bombardment due to synaptic activity, which acts as a major source of noise. Here, we investigate the effects of the noise filtering by synapses with various levels of realism on integrate-and-fire neuron dynamics. The noise input is modeled by white (for instantaneous synapses) or colored (for synapses with a finite relaxation time) noise. Analytical results for the modulation of firing probability in response to an oscillatory input current are obtained by expanding a Fokker-Planck equation for small parameters of the problem - when both the amplitude of the modulation is small compared to the background firing rate and the synaptic time constant is small compared to the membrane time constant. We report here the detailed calculations showing that if a synaptic decay time constant is included in the synaptic current model, the firing-rate modulation of the neuron due to an oscillatory input remains finite in the high-frequency limit with no phase lag. In addition, we characterize the low-frequency behavior and the behavior of the high-frequency limit for intermediate decay times. We also characterize the effects of introducing a rise time to the synaptic currents and the presence of several synaptic receptors with different kinetics. In both cases, we determine, using numerical simulations, an effective decay time constant that describes the neuronal response completely.

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100. Integrate-and-Fire Neurons Driven by Correlated Stochastic Input

Emilio Salinas, Terrence J. Sejnowski

https://doi.org/10.1162/089976602320264024 | pp. 2111-2155

Neurons are sensitive to correlations among synaptic inputs. However, analytical models that explicitly include correlations are hard to solve analytically, so their influence on a neuron's response has been difficult to ascertain. To gain some intuition on this problem, we studied the firing times of two simple integrate-and-fire model neurons driven by a correlated binary variable that represents the total input current. Analytic expressions were obtained for the average firing rate and coefficient of variation (a measure of spike-train variability) as functions of the mean, variance, and correlation time of the stochastic input. The results of computer simulations were in excellent agreement with these expressions. In these models, an increase in correlation time in general produces an increase in both the average firing rate and the variability of the output spike trains. However, the magnitude of the changes depends differentially on

the relative values of the input mean and variance: the increase in firing rate is higher when the variance is large relative to the mean, whereas the increase in variability is higher when the variance is relatively small. In addition, the firing rate always tends to a finite limit value as the correlation time increases toward infinity, whereas the coefficient of variation typically diverges. These results suggest that temporal correlations may play a major role in determining the variability as well as the intensity of neuronal spike trains.

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101. Preintegration Lateral Inhibition Enhances Unsupervised Learning

Michael Spratling, Mark H. Johnson

https://doi.org/10.1162/089976602320264033 | pp. 2157-2179

A large and influential class of neural network architectures uses postintegration lateral inhibition as a mechanism for competition. We argue that these algorithms are computationally deficient in that they fail to generate, or learn, appropriate perceptual representations under certain circumstances. An alternative neural network architecture is presented here in which nodes compete for the right to receive inputs rather than for the right to generate outputs. This form of competition, implemented through preintegration lateral inhibition, does provide appropriate coding properties and can be used to learn such representations efficiently. Furthermore, this architecture is consistent with both neuroanatomical and neurophysiological data. We thus argue that preintegration lateral inhibition has computational advantages over conventional neural network architectures while remaining equally biologically plausible.

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102. On Optimality in Auditory Information Processing

Mattias F. Karlsson, John W. Robinson

https://doi.org/10.1162/089976602320264042 | pp. 2181-2200

We study limits for the detection and estimation of weak sinusoidal signals in the primary part of the mammalian auditory system using a stochastic Fitzhugh-Nagumo model and an action-recovery model for synaptic depression. Our overall model covers the chain from a hair cell to a point just after the synaptic connection with a cell in the cochlear nucleus. The information processing performance of the system is evaluated using so-called phi-divergences from statistics that quantify "dissimilarity" between probability measures and are intimately related to a number of fundamental limits in statistics and information theory (IT). We show that there exists a set of parameters that can optimize several important phi-divergences simultaneously and that this set corresponds to a constant quiescent firing rate (QFR) of the spiral ganglion neuron. The optimal value of the QFR is frequency dependent but is essentially independent of the amplitude of the signal (for small amplitudes). Consequently, optimal processing according to several standard IT criteria can be accomplished for this model if and only if the parameters are "tuned" to values that correspond to one and the same QFR. This offers a new explanation for the QFR and can provide new insight into the role played by several other parameters of the peripheral auditory system.

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103. Computational Capacity of an Odorant Discriminator: The Linear Separability of Curves

Nestor Caticha, José Esquicha-Tejada, Doron Lancet, Eytan Domany

https://doi.org/10.1162/089976602320264051 | pp. 2201-2220

We introduce and study an artificial neural network inspired by the probabilistic receptor affinity distribution model of olfaction. Our system consists of N sensory neurons whose outputs converge on a single processing linear threshold element. The system's aim is to model discrimination of a single target odorant from a large number p of background odorants within a range of odorant concentrations. We show that this is possible provided p does not exceed a critical value p(c) and calculate the critical capacity alpha(c) = p(c)/N. The critical capacity depends on the range of concentrations in which the discrimination is to be accomplished. If the olfactory bulb may be thought of as a collection of such processing elements, each responsible for the discrimination of a single odorant, our study provides a quantitative analysis of the potential computational properties of the olfactory bulb. The mathematical formulation of the problem we consider is one of determining the capacity for linear separability of continuous curves, embedded in a large-dimensional space. This is accomplished here by a numerical study, using a method that signals whether the discrimination task is realizable, together with a finite-size scaling analysis.

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104. Mixture of Experts Classification Using a Hierarchical Mixture Model

Michalis K. Titsias, Aristidis Likas

https://doi.org/10.1162/089976602320264060 | pp. 2221-2244

A three-level hierarchical mixture model for classification is presented that models the following data generation process: (1) the data are generated by a finite number of sources (clusters), and (2) the generation mechanism of each source assumes the existence of individual internal class-labeled sources (subclusters of the external cluster). The model estimates the posterior probability of class membership similar to a mixture of experts classifier. In order to learn the parameters of the model, we have developed a general training approach based on maximum likelihood that results in two efficient training algorithms. Compared to other classification mixture models, the proposed hierarchical model exhibits several advantages and provides improved classification performance as indicated by the experimental results.

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105. On the Emergence of Rules in Neural Networks

Stephen José Hanson, Michiro Negishi

https://doi.org/10.1162/089976602320264079 | pp. 2245-2268

A simple associationist neural network learns to factor abstract rules (i.e., grammars) from sequences of arbitrary input symbols by inventing abstract representations that accommodate unseen symbol sets as well as unseen but similar grammars. The neural network is shown to have the ability to transfer grammatical knowledge to both new symbol vocabularies and new grammars. Analysis of the state-space shows that the network learns generalized abstract structures of the input and is not simply memorizing the input strings. These representations are context sensitive, hierarchical, and based on the state variable of the finite-state machines that the neural network has learned. Generalization to new symbol sets or grammars arises from the spatial nature of the internal representations used by the network, allowing new symbol sets to be encoded close to symbol sets that have already been learned in the hidden unit space of the network. The results are counter to the arguments that learning algorithms based on weight adaptation after each exemplar presentation (such as the long term potentiation found in the mammalian nervous system) cannot in principle extract symbolic knowledge from positive examples as prescribed by prevailing human linguistic

theory and evolutionary psychology.

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106. Information-Geometric Measure for Neural Spikes

Hiroyuki Nakahara, ■■■-■■■ ■■■■■

https://doi.org/10.1162/08997660260293238 | pp. 2269-2316

This study introduces information-geometric measures to analyze neural firing patterns by taking not only the second-order but also higher-order interactions among neurons into account. Information geometry provides useful tools and concepts for this purpose, including the orthogonality of coordinate parameters and the Pythagoras relation in the Kullback-Leibler divergence. Based on this orthogonality, we show a novel method for analyzing spike firing patterns by decomposing the interactions of neurons of various orders. As a result, purely pairwise, triple-wise, and higher-order interactions are singled out. We also demonstrate the benefits of our proposal by using several examples.

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107. Optimal Short-Term Population Coding: When Fisher Information Fails

Matthias Bethge, David Rotermund, Klaus Pawelzik

https://doi.org/10.1162/08997660260293247 | pp. 2317-2351

Efficient coding has been proposed as a first principle explaining neuronal response properties in the central nervous system. The shape of optimal codes, however, strongly depends on the natural limitations of the particular physical system. Here we investigate how optimal neuronal encoding strategies are influenced by the finite number of neurons N (place constraint), the limited decoding time window length T (time constraint), the maximum neuronal firing rate f(max) (power constraint), and the maximal average rate (f)(max) (energy constraint). While Fisher information provides a general lower bound for the mean squared error of unbiased signal reconstruction, its use to characterize the coding precision is limited. Analyzing simple examples, we illustrate some typical pitfalls and thereby show that Fisher information provides a valid measure for the precision of a code only if the dynamic range (f(min)T, f(max)T) is sufficiently large. In particular, we demonstrate that the optimal width of gaussian tuning curves depends on the available decoding time T. Within the broader class of unimodal tuning functions, it turns out that the shape of a Fisher-optimal coding scheme is not unique. We solve this ambiguity by taking the minimum mean square error into account, which leads to flat tuning curves. The tuning width, however, remains to be determined by energy constraints rather than by the principle of efficient coding.

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108. Coupling an aVLSI Neuromorphic Vision Chip to a Neurotrophic Model of Synaptic Plasticity: The Development of Topography

Terry Elliott, Jörg Krämer

https://doi.org/10.1162/08997660260293256 | pp. 2353-2370

We couple a previously studied, biologically inspired neurotrophic model of activity-dependent competitive synaptic plasticity and neuronal development to a neuromorphic retina chip. Using this system, we examine the development and refinement of a topographic mapping between an array of afferent neurons (the retinal ganglion cells) and an array of target neurons. We find that the plasticity model can indeed drive topographic

refinement in the presence of afferent activity patterns generated by a real-world device. We examine the resilience of the developing system to the presence of high levels of noise by adjusting the spontaneous firing rate of the silicon neurons.

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109. Hebbian Imprinting and Retrieval in Oscillatory Neural Networks

Silvia Scarpetta, L. Zhaoping, John Hertz

https://doi.org/10.1162/08997660260293265 | pp. 2371-2396

We introduce a model of generalized Hebbian learning and retrieval in oscillatory neural networks modeling cortical areas such as hippocampus and olfactory cortex. Recent experiments have shown that synaptic plasticity depends on spike timing, especially on synapses from excitatory pyramidal cells, in hippocampus, and in sensory and cerebellar cortex. Here we study how such plasticity can be used to form memories and input representations when the neural dynamics are oscillatory, as is common in the brain (particularly in the hippocampus and olfactory cortex). Learning is assumed to occur in a phase of neural plasticity, in which the network is clamped to external teaching signals. By suitable manipulation of the nonlinearity of the neurons or the oscillation frequencies during learning, the model can be made, in a retrieval phase, either to categorize new inputs or to map them, in a continuous fashion, onto the space spanned by the imprinted patterns. We identify the first of these possibilities with the function of olfactory cortex and the second with the observed response characteristics of place cells in hippocampus. We investigate both kinds of networks analytically and by computer simulations, and we link the models with experimental findings, exploring, in particular, how the spike timing dependence of the synaptic plasticity constrains the computational function of the network and vice versa.

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110. A New Discriminative Kernel from Probabilistic Models

Koji Tsuda, Motoaki Kawanabe, Gunnar Rätsch, Sören Sonnenburg, Klaus■Robert Müller

https://doi.org/10.1162/08997660260293274 | pp. 2397-2414

Recently, Jaakkola and Haussler (1999) proposed a method for constructing kernel functions from probabilistic models. Their so-called Fisher kernel has been combined with discriminative classifiers such as support vector machines and applied successfully in, for example, DNA and protein analysis. Whereas the Fisher kernel is calculated from the marginal log-likelihood, we propose the TOP kernel derived; from tangent vectors of posterior log-odds. Furthermore, we develop a theoretical framework on feature extractors from probabilistic models and use it for analyzing the TOP kernel. In experiments, our new discriminative TOP kernel compares favorably to the Fisher kernel.

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111. Factorial Hidden Markov Models and the Generalized Backfitting Algorithm

Robert A. Jacobs, Wenxin Jiang, Martin A. Tanner

https://doi.org/10.1162/08997660260293283 | pp. 2415-2437

Previous researchers developed new learning architectures for sequential data by extending conventional hidden Markov models through the use of distributed state representations. Although exact inference and

parameter estimation in these architectures is computationally intractable, Ghahramani and Jordan (1997) showed that approximate inference and parameter estimation in one such architecture, factorial hidden Markov models (FHMMs), is feasible in certain circumstances. However, the learning algorithm proposed by these investigators, based on variational techniques, is difficult to understand and implement and is limited to the study of real-valued data sets. This chapter proposes an alternative method for approximate inference and parameter estimation in FHMMs based on the perspective that FHMMs are a generalization of a well-known class of statistical models known as generalized additive models (GAMs; Hastie & Tibshirani, 1990). Using existing statistical techniques for GAMs as a guide, we have developed the generalized backfitting algorithm. This algorithm computes customized error signals for each hidden Markov chain of an FHMM and then trains each chain one at a time using conventional techniques from the hidden Markov models literature. Relative to previous perspectives on FHMMs, we believe that the viewpoint taken here has a number of advantages. First, it places FHMMs on firm statistical foundations by relating them to a class of models that are well studied in the statistics community, yet it generalizes this class of models in an interesting way. Second, it leads to an understanding of how FHMMs can be applied to many different types of time-series data, including Bernoulli and multinomial data, not just data that are real valued. Finally, it leads to an effective learning procedure for FHMMs that is easier to understand and easier to implement than existing learning procedures. Simulation results suggest that FHMMs trained with the generalized backfitting algorithm are a practical and powerful tool for analyzing sequential data.

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112. Bayesian Model Assessment and Comparison Using Cross-Validation Predictive Densities

Aki Vehtari, Jouko Lampinen

https://doi.org/10.1162/08997660260293292 | pp. 2439-2468

In this work, we discuss practical methods for the assessment, comparison, and selection of complex hierarchical Bayesian models. A natural way to assess the goodness of the model is to estimate its future predictive capability by estimating expected utilities. Instead of just making a point estimate, it is important to obtain the distribution of the expected utility estimate because it describes the uncertainty in the estimate. The distributions of the expected utility estimates can also be used to compare models, for example, by computing the probability of one model having a better expected utility than some other model. We propose an approach using cross-validation predictive densities to obtain expected utility estimates and Bayesian bootstrap to obtain samples from their distributions. We also discuss the probabilistic assumptions made and properties of two practical cross-validation methods, importance sampling and k-fold cross-validation. As illustrative examples, we use multilayer perceptron neural networks and gaussian processes with Markov chain Monte Carlo sampling in one toy problem and two challenging real-world problems.

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113. Robust Regression with Asymmetric Heavy-Tail Noise Distributions

Ichiro Takeuchi, Yoshua Bengio, Takafumi Kanamori

https://doi.org/10.1162/08997660260293300 | pp. 2469-2496

In the presence of a heavy-tail noise distribution, regression becomes much more difficult. Traditional robust regression methods assume that the noise distribution is symmetric, and they downweight the influence of so-called outliers. When the noise distribution is asymmetric, these methods yield biased regression

estimators. Motivated by data-mining problems for the insurance industry, we propose a new approach to robust regression tailored to deal with asymmetric noise distribution. The main idea is to learn most of the parameters of the model using conditional quantile estimators (which are biased but robust estimators of the regression) and to learn a few remaining parameters to combine and correct these estimators, to minimize the average squared error in an unbiased way. Theoretical analysis and experiments show the clear advantages of the approach. Results are on artificial data as well as insurance data, using both linear and neural network predictors.

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114. Many-Layered Learning

Paul E. Utgoff, David John Stracuzzi

https://doi.org/10.1162/08997660260293319 | pp. 2497-2529

We explore incremental assimilation of new knowledge by sequential learning. Of particular interest is how a network of many knowledge layers can be constructed in an on-line manner, such that the learned units represent building blocks of knowledge that serve to compress the overall representation and facilitate transfer. We motivate the need for many layers of knowledge, and we advocate sequential learning as an avenue for promoting the construction of layered knowledge structures. Finally, our novel STL algorithm demonstrates a method for simultaneously acquiring and organizing a collection of concepts and functions as a network from a stream of unstructured information.

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115. Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations

Wolfgang Maass, Thomas Natschläger, Henry Markram

https://doi.org/10.1162/089976602760407955 | pp. 2531-2560

A key challenge for neural modeling is to explain how a continuous stream of multimodal input from a rapidly changing environment can be processed by stereotypical recurrent circuits of integrate-and-fire neurons in real time. We propose a new computational model for real-time computing on time-varying input that provides an alternative to paradigms based on Turing machines or attractor neural networks. It does not require a task-dependent construction of neural circuits. Instead, it is based on principles of high-dimensional dynamical systems in combination with statistical learning theory and can be implemented on generic evolved or found recurrent circuitry. It is shown that the inherent transient dynamics of the high-dimensional dynamical system formed by a sufficiently large and heterogeneous neural circuit may serve as universal analog fading memory. Readout neurons can learn to extract in real time from the current state of such recurrent neural circuit information about current and past inputs that may be needed for diverse tasks. Stable internal states are not required for giving a stable output, since transient internal states can be transformed by readout neurons into stable target outputs due to the high dimensionality of the dynamical system. Our approach is based on a rigorous computational model, the liquid state machine, that, unlike Turing machines, does not require sequential transitions between well-defined discrete internal states. It is supported, as the Turing machine is, by rigorous mathematical results that predict universal computational power under idealized conditions, but for the biologically more realistic scenario of real-time processing of time-varying inputs. Our approach provides new perspectives for the interpretation of neural coding, the design of experiments and data analysis in neurophysiology, and the solution of problems in

robotics and neurotechnology.

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116. Universal Approximation of Multiple Nonlinear Operators by Neural Networks

Andrew D. Back, Tianping Chen

https://doi.org/10.1162/089976602760407964 | pp. 2561-2566

Recently, there has been interest in the observed capabilities of some classes of neural networks with fixed weights to model multiple nonlinear dynamical systems. While this property has been observed in simulations, open questions exist as to how this property can arise. In this article, we propose a theory that provides a possible mechanism by which this multiple modeling phenomenon can occur.

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117. Long-Term Reward Prediction in TD Models of the Dopamine System

Nathaniel D. Daw, David S. Touretzky

https://doi.org/10.1162/089976602760407973 | pp. 2567-2583

This article addresses the relationship between long-term reward predictions and slow-timescale neural activity in temporal difference (TD) models of the dopamine system. Such models attempt to explain how the activity of dopamine (DA) neurons relates to errors in the prediction of future rewards. Previous models have been mostly restricted to short-term predictions of rewards expected during a single, somewhat artificially defined trial. Also, the models focused exclusively on the phasic pause-and-burst activity of primate DA neurons; the neurons' slower, tonic background activity was assumed to be constant. This has led to difficulty in explaining the results of neurochemical experiments that measure indications of DA release on a slow timescale, results that seem at first glance inconsistent with a reward prediction model. In this article, we investigate a TD model of DA activity modified so as to enable it to make longer-term predictions about rewards expected far in the future. We show that these predictions manifest themselves as slow changes in the baseline error signal, which we associate with tonic DA activity. Using this model, we make new predictions about the behavior of the DA system in a number of experimental situations. Some of these predictions suggest new computational explanations for previously puzzling data, such as indications from microdialysis studies of elevated DA activity triggered by aversive events.

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118. Invariant Object Recognition in the Visual System with Novel Views of 3D Objects

Simon M. Stringer, Edmund T. Rolls

https://doi.org/10.1162/089976602760407982 | pp. 2585-2596

To form view-invariant representations of objects, neurons in the inferior temporal cortex may associate together different views of an object, which tend to occur close together in time under natural viewing conditions. This can be achieved in neuronal network models of this process by using an associative learning rule with a short-term temporal memory trace. It is postulated that within a view, neurons learn representations that enable them to generalize within variations of that view. When three-dimensional (3D) objects are rotated within small angles (up to, e.g., 30 degrees), their surface features undergo geometric

distortion due to the change of perspective. In this article, we show how trace learning could solve the problem of in-depth rotation-invariant object recognition by developing representations of the transforms that features undergo when they are on the surfaces of 3D objects. Moreover, we show that having learned how features on 3D objects transform geometrically as the object is rotated in depth, the network can correctly recognize novel 3D variations within a generic view of an object composed of a new combination of previously learned features. These results are demonstrated in simulations of a hierarchical network model (VisNet) of the visual system that show that it can develop representations useful for the recognition of 3D objects by forming perspective-invariant representations to allow generalization within a generic view.

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119. Dynamical Working Memory and Timed Responses: The Role of Reverberating Loops in the Olivo-Cerebellar System

Werner M. Kistler, Chris I. De Zeeuw

https://doi.org/10.1162/089976602760407991 | pp. 2597-2626

This article explores dynamical properties of the olivo-cerebellar system that arise from the specific wiring of inferior olive (IO), cerebellar cortex, and deep cerebellar nuclei (DCN). We show that the irregularity observed in the firing pattern of the IO neurons is not necessarily produced by noise but can instead be the result of a purely deterministic network effect. We propose that this effect can serve as a dynamical working memory or as a neuronal clock with a characteristic timescale of about 100 ms that is determined by the slow calcium dynamics of IO and DCN neurons. This concept provides a novel explanation of how the cerebellum can solve timing tasks on a timescale that is two orders of magnitude longer than the millisecond timescale usually attributed to neuronal dynamics. One of the key ingredients of our model is the observation that due to postinhibitory rebound, DCN neurons can be driven by GABAergic ("inhibitory") input from cerebellar Purkinje cells. Topographic projections from the DCN to the IO form a closed reverberating loop with an overall synaptic transmission delay of about 100 ms that is in resonance with the intrinsic oscillatory properties of the inferior olive. We use a simple time-discrete model based on McCulloch-Pitts neurons in order to investigate in a first step some of the fundamental properties of a network with delayed reverberating projections. The macroscopic behavior is analyzed by means of a mean-field approximation. Numerical simulations, however, show that the microscopic dynamics has a surprisingly rich structure that does not show up in a mean-field description. We have thus performed extensive numerical experiments in order to quantify the ability of the network to serve as a dynamical working memory and its vulnerability by noise. In a second step, we develop a more realistic conductance-based network model of the inferior olive consisting of about 20 multicompartment neurons that are coupled by gap junctions and receive excitatory and inhibitory synaptic input via AMPA and GABAergic synapses. The simulations show that results for the time-discrete model hold true in a time-continuous description.

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120. Selectively Grouping Neurons in Recurrent Networks of Lateral Inhibition

Xiaohui Xie, Richard H. R. Hahnloser, H. Sebastian Seung

https://doi.org/10.1162/089976602760408008 | pp. 2627-2646

Winner-take-all networks have been proposed to underlie many of the brain's fundamental computational abilities. However, not much is known about how to extend the grouping of potential winners in these

networks beyond single neuron or uniformly arranged groups of neurons. We show that competition between arbitrary groups of neurons can be realized by organizing lateral inhibition in linear threshold networks. Given a collection of potentially overlapping groups (with the exception of some degenerate cases), the lateral inhibition results in network dynamics such that any permitted set of neurons that can be coactivated by some input at a stable steady state is contained in one of the groups. The information about the input is preserved in this operation. The activity level of a neuron in a permitted set corresponds to its stimulus strength, amplified by some constant. Sets of neurons that are not part of a group cannot be coactivated by any input at a stable steady state. We analyze the storage capacity of such a network for random groups--the number of random groups the network can store as permitted sets without creating too many spurious ones. In this framework, we calculate the optimal sparsity of the groups (maximizing group entropy). We find that for dense inputs, the optimal sparsity is unphysiologically small. However, when the inputs and the groups are equally sparse, we derive a more plausible optimal sparsity. We believe our results are the first steps toward attractor theories in hybrid analog-digital networks.

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121. An Unsupervised Ensemble Learning Method for Nonlinear Dynamic State-Space Models

Harri Valpola, Juha Karhunen

https://doi.org/10.1162/089976602760408017 | pp. 2647-2692

A Bayesian ensemble learning method is introduced for unsupervised extraction of dynamic processes from noisy data. The data are assumed to be generated by an unknown nonlinear mapping from unknown factors. The dynamics of the factors are modeled using a nonlinear state-space model. The nonlinear mappings in the model are represented using multilayer perceptron networks. The proposed method is computationally demanding, but it allows the use of higher-dimensional nonlinear latent variable models than other existing approaches. Experiments with chaotic data show that the new method is able to blindly estimate the factors and the dynamic process that generated the data. It clearly outperforms currently available nonlinear prediction techniques in this very difficult test problem.

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122. Data-Reusing Recurrent Neural Adaptive Filters

Danilo P. Mandic

https://doi.org/10.1162/089976602760408026 | pp. 2693-2707

A class of data-reusing learning algorithms for real-time recurrent neural networks (RNNs) is analyzed. The analysis is undertaken for a general sigmoid nonlinear activation function of a neuron for the real time recurrent learning training algorithm. Error bounds and convergence conditions for such data-reusing algorithms are provided for both contractive and expansive activation functions. The analysis is undertaken for various configurations that are generalizations of a linear structure infinite impulse response adaptive filter.

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123. Training a Single Sigmoidal Neuron Is Hard

Ji■■■ Š■■ma

https://doi.org/10.1162/089976602760408035 | pp. 2709-2728

We first present a brief survey of hardness results for training feedforward neural networks. These results are then completed by the proof that the simplest architecture containing only a single neuron that applies a sigmoidal activation function sigma: kappa --> [alpha, beta], satisfying certain natural axioms (e.g., the standard (logistic) sigmoid or saturated-linear function), to the weighted sum of n inputs is hard to train. In particular, the problem of finding the weights of such a unit that minimize the quadratic training error within (beta - alpha)(2) or its average (over a training set) within 5(beta - alpha)(2)/ (12n) of its infimum proves to be NP-hard. Hence, the well-known backpropagation learning algorithm appears not to be efficient even for one neuron, which has negative consequences in constructive learning.

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124. Two Timescale Analysis of the Alopex Algorithm for Optimization

P. S. Sastry, M. Magesh, K. P. Unnikrishnan

https://doi.org/10.1162/089976602760408044 | pp. 2729-2750

Alopex is a correlation-based gradient-free optimization technique useful in many learning problems. However, there are no analytical results on the asymptotic behavior of this algorithm. This article presents a new version of Alopex that can be analyzed using techniques of two timescale stochastic approximation method. It is shown that the algorithm asymptotically behaves like a gradient-descent method, though it does not need (or estimate) any gradient information. It is also shown, through simulations, that the algorithm is quite effective.

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125. A New Color 3D SFS Methodology Using Neural-Based Color Reflectance Models and Iterative Recursive Method

Siu-Yeung Cho, Tommy W. S. Chow

https://doi.org/10.1162/089976602760408053 | pp. 2751-2789

In this article, a new methodology for color shape from shading (SFS) problem is proposed. The problem of color SFS refers to the well-known fact that most real objects usually contain mixtures of diffuse and specular color reflections and are affected by the multicolored interreflection under unknown reflectivity. In this article, these limitations are addressed, and a new color SFS methodology is proposed. The proposed approach focuses on two main parts. First, a generalized neural-based color reflectance model is developed. Second, an iterative recursive method is developed to reconstruct a multicolor 3D surface. Experimental results on synthetic-colored objects and real-colored objects were performed to demonstrate the performance of the proposed methodology.

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126. On Different Facets of Regularization Theory

Zhe Chen, S. Haykin

https://doi.org/10.1162/089976602760805296 | pp. 2791-2846

This review provides a comprehensive understanding of regularization theory from different perspectives, emphasizing smoothness and simplicity principles. Using the tools of operator theory and Fourier analysis, it is shown that the solution of the classical Tikhonov regularization problem can be derived from the

regularized functional defined by a linear differential (integral) operator in the spatial (Fourier) domain. State-of-the-art research relevant to the regularization theory is reviewed, covering Occam's razor, minimum length description, Bayesian theory, pruning algorithms, informational (entropy) theory, statistical learning theory, and equivalent regularization. The universal principle of regularization in terms of Kolmogorov complexity is discussed. Finally, some prospective studies on regularization theory and beyond are suggested.

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127. Notes on Bell-Sejnowski PDF-Matching Neuron

Simone Fiori

https://doi.org/10.1162/089976602760805304 | pp. 2847-2855

This article investigated the behavior of a single-input, single-unit neuron model of the Bell-Sejnowski class, which learn through the maximum-entropy principle, in order to understand its probability density function and matching ability.

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128. Biophysiologically Plausible Implementations of the Maximum Operation

Angela J. Yu, Martin A. Giese, Tomaso Poggio

https://doi.org/10.1162/089976602760805313 | pp. 2857-2881

Visual processing in the cortex can be characterized by a predominantly hierarchical architecture, in which specialized brain regions along the processing pathways extract visual features of increasing complexity, accompanied by greater invariance in stimulus properties such as size and position. Various studies have postulated that a nonlinear pooling function such as the maximum (MAX) operation could be fundamental in achieving such selectivity and invariance. In this article, we are concerned with neurally plausible mechanisms that may be involved in realizing the MAX operation. Different canonical models are proposed, each based on neural mechanisms that have been previously discussed in the context of cortical processing. Through simulations and mathematical analysis, we compare the performance and robustness of these mechanisms. We derive experimentally verifiable predictions for each model and discuss the relevant physiological considerations.

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129. Self-Regulation Mechanism of Temporally Asymmetric Hebbian Plasticity

Narihisa Matsumoto, Masato Okada

https://doi.org/10.1162/089976602760805322 | pp. 2883-2902

Recent biological experimental findings have shown that synaptic plasticity depends on the relative timing of the pre- and postsynaptic spikes. This determines whether long-term potentiation (LTP) or long-term depression (LTD) is induced. This synaptic plasticity has been called temporally asymmetric Hebbian plasticity (TAH). Many authors have numerically demonstrated that neural networks are capable of storing spatiotemporal patterns. However, the mathematical mechanism of the storage of spatiotemporal patterns is still unknown, and the effect of LTD is particularly unknown. In this article, we employ a simple neural

network model and show that interference between LTP and LTD disappears in a sparse coding scheme. On the other hand, the covariance learning rule is known to be indispensable for the storage of sparse patterns. We also show that TAH has the same qualitative effect as the covariance rule when spatiotemporal patterns are embedded in the network.

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130. Associative Memory with Dynamic Synapses

Lovorka Pantic, Joaqu■■n J. Torres, Hilbert J. Kappen, Stan Gielen

https://doi.org/10.1162/089976602760805331 | pp. 2903-2923

We have examined a role of dynamic synapses in the stochastic Hopfield-like network behavior. Our results demonstrate an appearance of a novel phase characterized by quick transitions from one memory state to another. The network is able to retrieve memorized patterns corresponding to classical ferromagnetic states but switches between memorized patterns with an intermittent type of behavior. This phenomenon might reflect the flexibility of real neural systems and their readiness to receive and respond to novel and changing external stimuli.

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131. Adaptive Spatiotemporal Receptive Field Estimation in the Visual Pathway

Garrett B. Stanley

https://doi.org/10.1162/089976602760805340 | pp. 2925-2946

The encoding properties of the visual pathway are under constant control from mechanisms of adaptation and systems-level plasticity. In all but the most artificial experimental conditions, these mechanisms serve to continuously modulate the spatial and temporal receptive field (RF) dynamics. Conventional reverse-correlation techniques designed to capture spatiotemporal RF properties assume invariant stimulus-response relationships over experimental trials and are thus limited in their applicability to more natural experimental conditions. Presented here is an approach to tracking time-varying encoding dynamics in the early visual pathway based on adaptive estimation of the spatiotemporal RF in the time domain. Simulations and experimental data from the lateral geniculate nucleus reveal that subtle features of encoding properties can be captured by the adaptive approach that would otherwise be undetected. Capturing the role of dynamically varying encoding mechanisms is vital to our understanding of vision on the natural setting, where there is absence of a true steady state.

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132. Global Convergence Rate of Recurrently Connected Neural Networks

Tianping Chen, Wenlian Lu, ■■■-■■■ ■■■■■

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We discuss recurrently connected neural networks, investigating their global exponential stability (GES). Some sufficient conditions for a class of recurrent neural networks belonging to GES are given. Sharp convergence rate is given too.

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133. Locality of Global Stochastic Interaction in Directed Acyclic Networks

Nihat Ay

https://doi.org/10.1162/089976602760805368 | pp. 2959-2980

The hypothesis of invariant maximization of interaction (IMI) is formulated within the setting of random fields. According to this hypothesis, learning processes maximize the stochastic interaction of the neurons subject to constraints. We consider the extrinsic constraint in terms of a fixed input distribution on the periphery of the network. Our main intrinsic constraint is given by a directed acyclic network structure. First mathematical results about the strong relation of the local information flow and the global interaction are stated in order to investigate the possibility of controlling IMI optimization in a completely local way. Furthermore, we discuss some relations of this approach to the optimization according to Linsker's Infomax principle.

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134. On Unique Representations of Certain Dynamical Systems Produced by Continuous-Time Recurrent Neural Networks

Masahiro Kimura

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This article extends previous mathematical studies on elucidating the redundancy for describing functions by feedforward neural networks (FNNs) to the elucidation of redundancy for describing dynamical systems (DSs) by continuous-time recurrent neural networks (RNNs). In order to approximate a DS on R(n) using an RNN with n visible units, an n-dimensional affine neural dynamical system (A-NDS) can be used as the DS actually produced by the above RNN under an affine map from its visible state-space R(n) to its hidden state-space. Therefore, we consider the problem of clarifying the redundancy for describing A-NDSs by RNNs and affine maps. We clarify to what extent a pair of an RNN and an affine map is uniquely determined by its corresponding A-NDS and also give a nonredundant sufficient search set for the DS approximation problem based on A-NDS.

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135. Descartes' Rule of Signs for Radial Basis Function Neural Networks

Michael Schmitt

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We establish versions of Descartes' rule of signs for radial basis function (RBF) neural networks. The RBF rules of signs provide tight bounds for the number of zeros of univariate networks with certain parameter restrictions. Moreover, they can be used to infer that the Vapnik-Chervonenkis (VC) dimension and pseudodimension of these networks are no more than linear. This contrasts with previous work showing that RBF neural networks with two or more input nodes have superlinear VC dimension. The rules also give rise to lower bounds for network sizes, thus demonstrating the relevance of network parameters for the complexity of computing with RBF neural networks.

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136. Approximation Bounds for Some Sparse Kernel Regression Algorithms

Tong Zhang

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Gaussian processes have been widely applied to regression problems with good performance. However, they can be computationally expensive. In order to reduce the computational cost, there have been recent studies on using sparse approximations in gaussian processes. In this article, we investigate properties of certain sparse regression algorithms that approximately solve a gaussian process. We obtain approximation bounds and compare our results with related methods.

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137. Index

(authors unknown)

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